CHAPTER 4

VECTOR SPACES IN PHYSICS

INTRODUCTION

So far, the concepts of length and angle have been missing from our discussion of vector spaces. Thus there have been few applications to physics. In this chapter the structure of the vector space will be enormously enriched by the addition of a numerical function, the *inner* (or scalar) product, in terms of which the length of a vector and the angle between two vectors can be defined. Vector spaces in which an inner product has been defined are called *inner-product spaces*. The study of inner-product spaces will enable us to make a real juncture with physics. Accordingly, in the latter sections of this chapter we shall discuss several examples from classical and quantum physics.

4.1 THE INNER PRODUCT

In Section 1.3 we defined the scalar product of two ordinary vectors in real three-dimensional space. In the notation used there, we wrote the scalar product of the vectors $\mathbf{x} = (x_1, x_2, x_3)$ and $\mathbf{y} = (y_1, y_2, y_3)$ as

$$\mathbf{x}\cdot\mathbf{y} \equiv \sum_{i=1}^{3} x_i y_i \,.$$

In this notation the length of the vector $\mathbf{x}$ is denoted by $|\,\mathbf{x}\,|$ and is given by the quantity

$$|\,\mathbf{x}\,| \equiv (\mathbf{x}\cdot\mathbf{x})^{1/2} = \left(\sum_{i=1}^{3} x_i^2\right)^{1/2} .$$

The angle θ between two vectors $\mathbf{x}$ and $\mathbf{y}$ is given by

$$\theta = \cos^{-1}\left[\frac{\mathbf{x}\cdot\mathbf{y}}{|\,\mathbf{x}\,|\;|\,\mathbf{y}\,|}\right] .$$

In abstract vector spaces we shall denote the length, or magnitude, of a vector x by the symbol $||\,x\,||$, in order to reserve single vertical bars for the absolute value of a complex number. It is customary in vector-space theory to refer to the scalar product as the *inner* product, and to write the inner product of two vectors, x and y, as (x, y). In this notation, the length of the vector x is written

$$||\,x\,|| \equiv (x, x)^{1/2} , \tag{4.1}$$

and the angle θ between two vectors is written

$$\theta = \cos^{-1}\left[\frac{(x, y)}{||x||\,||y||}\right]. \tag{4.2}$$

As long as we refer to real three-dimensional vector spaces, we can define the quantity (x, y) just as we defined $\mathbf{x}\cdot\mathbf{y}$. That is, if $x = (\xi_1, \xi_2, \xi_3)$ and $y = (\eta_1, \eta_2, \eta_3)$, then

$$(x, y) \equiv \sum_{i=1}^{3} \xi_i \eta_i . \tag{4.3}$$

The generalization to real n-dimensional vector spaces is also straightforward—we merely extend the sum over i to n. However, if we try to retain this definition of the inner product for a *complex* vector space, we immediately encounter difficulties. For example, consider the "length" of the one-dimensional vector ix. The above definition tells us that its length is

$$||ix|| = (ix, ix)^{1/2} = (-x^2)^{1/2},$$

which is imaginary if x is real. But a distance function must be real-valued. We could skirt the difficulty by restricting our attention to real vector spaces, but only at the price of losing relevance to much of physics. In particular, in quantum theory the underlying field is not the real field, but the field of complex numbers.

There is a simple, but seemingly artificial, remedy for this situation. Let x and y be vectors with *complex* components $(\xi_1, \xi_2, \cdots, \xi_n)$ and $(\eta_1, \eta_2, \cdots, \eta_n)$ referred to some basis. We shall not define the inner product of these vectors by Eq. (4.3), but rather by

$$(x, y) \equiv \sum_{i=1}^{n} \xi_i^* \eta_i . \tag{4.4}$$

The asterisk (*) denotes complex conjugation of ξ_i, which is a number in the complex field.* With this definition of the inner product, the length of the vector x is

$$||x|| \equiv (x, x)^{1/2} = \left(\sum_{i=1}^{n} \xi_i^* \xi_i\right)^{1/2} = \left(\sum_{i=1}^{n} |\xi_i|^2\right)^{1/2}, \tag{4.5}$$

where $|\xi_i|$ is the absolute value of ξ_i. Since $|\xi_i|^2 \geq 0$, $||x||$ is always real. Moreover, if the vector space is real, $\xi_i = \xi_i^*$, and this definition reduces to the familiar one of Eq. (4.3).

For a complex vector space, the inner product is not symmetrical as it is in a real vector space. That is,

$$(x, y) \neq (y, x),$$

* If $z = a + ib$, where $i \equiv \sqrt{-1}$, then $z^* \equiv a - ib$. The absolute value of z is the quantity $|z| \equiv (zz^*)^{1/2} = (a^2 + b^2)^{1/2}$.

but rather

$$(x, y) = (y, x)^* . \tag{4.6}$$

Nevertheless, this definition will work out satisfactorily.

We now give the formal definition of an inner-product space.

Definition 4.1. An *inner product* in a (real or complex) vector space is a scalar-valued function of the ordered pair of vectors x and y, such that

1. $(x, y) = (y, x)^*$;
2. $(\alpha x + \beta y, z) = \alpha^*(x, z) + \beta^*(y, z)$, where α and β are scalars in the field underlying vector space;
3. $(x, x) \geq 0$ for any x; $(x, x) = 0$ if and only if $x = 0$.

An *inner-product space* is a vector space with an inner product. A real vector space with an inner product, that is, a real inner-product space, is called a *Euclidean* space; a complex inner-product space is called a *unitary* space.

In a real vector space, the complex conjugate in (1) and (2) adds nothing and may be ignored. In either case, real or complex, (1) implies that (x, x) is real, so that the inequality in (3) makes sense. The quantity $(x, x)^{1/2} \equiv ||x||$ is often referred to as the *norm* of the vector x. Note that

$$||ax|| = (ax, ax)^{1/2} = [a^*a(x, x)]^{1/2} = |a| \cdot ||x|| .$$

It follows from (1) and (2) that $(x, \alpha y) = \alpha(x, y)$.

Requirement (3) says that the inner product is a *positive-definite* form—it is *positive* unless x is *definitely* the zero vector, in which case it is *definitely* zero. As we proceed, the reader will see that certain theorems can be proved without reference to this property.

Examples of an Inner Product.

1. As we have seen, if

$$x = (\xi_1, \cdots, \xi_n) \qquad \text{and} \qquad y = (\eta_1, \cdots, \eta_n) ,$$

then

$$(x, y) \equiv \sum_{i=1}^{n} \xi_i^* \eta_i$$

satisfies all the axioms of an inner product.

2. If $x(t)$ and $y(t)$ are polynomials in the complex vector space P, defined on the closed interval $[0, 1]$, then an inner product is

$$(x, y) \equiv \int_0^1 x(t)^* y(t)\, dt;$$

3. If $x = (\xi_1, \cdots, \xi_n)$ and $y = (\eta_1, \cdots, \eta_n)$ are column vectors over a real field, and P is a diagonal matrix whose diagonal elements are real positive numbers,

then we may define an inner product as

$$(x, y) \equiv \bar{x}Py = (\xi_1, \cdots, \xi_n)\begin{bmatrix} P_{11} & & 0 \\ & \ddots & \\ 0 & & P_{nn} \end{bmatrix}\begin{pmatrix} \eta_1 \\ \vdots \\ \eta_n \end{pmatrix} = \sum_{i=1}^{n} \xi_i P_{ii} \eta_i .$$

If x is a column vector, $\bar{x}$ denotes its transpose, a row vector.

4. If $x = (\xi_1, \xi_2, \xi_3, \xi_4)$ and $y = (\eta_1, \eta_2, \eta_3, \eta_4)$ are column four-vectors over a real field, and

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix},$$

then the quantity $(x, y) \equiv \bar{x}My = \xi_1 n_1 + \xi_2 n_2 + \xi_3 n_3 - \xi_4 n_4$ satisfies requirements (1) and (2) for an inner product, but not (3)—it is not positive-definite. Thus the entity $\bar{x}My$ is not an inner product. The quantity $\bar{x}My$ plays an important role in the theory of special relativity; M is called the *Lorentz metric*.

4.2 ORTHOGONALITY AND COMPLETENESS

Now that the concepts of length and angle are defined, the theory of vector spaces becomes more immediately relevant to physics. One of the central notions of mathematical physics is that of complete orthogonal sets of vectors or functions. In this section we lay the groundwork for the theory of such sets.

Definition 4.2. Two vectors x and y are *orthogonal* if and only if $(x, y) = 0$.

Two vectors in a three-dimensional space are orthogonal if the angle between them is 90°; our definition is simply a generalization of this idea for n-dimensional—or even infinite-dimensional—spaces. As we might expect, if $(x, y) = 0$, then $(x, y) = (y, x)^* = 0$, so that $(y, x) = 0$ as well. Thus, although the inner product is unsymmetric, orthogonality is a symmetric relation. Note that the zero vector is orthogonal to every vector.

Definition 4.3. A set of vectors $\{x_1, x_2, \cdots\}$ is *orthonormal* if $(x_i, x_j) = \delta_{ij}$, for all i and j.

That is, a set of vectors is orthonormal if each vector is *orthogonal* to all others in the set and is *normalized* to unit length. A vector x may be normalized by dividing by its length to form the new vector $x/||x||$. One orthonormal set of vectors is the set of three unit vectors, $\{\mathbf{e}_i\}$, for the Cartesian space introduced in Section 1.3.

Definition 4.4. In a finite-dimensional vector space, an orthonormal set is *complete* if it is not contained in any larger orthonormal set.

We now prove several basic theorems involving these concepts.

Theorem 4.1. An orthonormal set is linearly independent. (Hence any orthonormal set of n vectors is a basis for an n-dimensional vector space.)

Proof. If $\{x_1, \cdots, x_n\}$ is any finite orthonormal set, then for any j,

$$\sum_i \alpha_i x_i = 0 \Longrightarrow 0 = \left(x_j, \sum_i \alpha_i x_i\right)$$
$$= \sum_i \alpha_i (x_j, x_i) = \sum_i \alpha_i \delta_{ij} = \alpha_j. \quad \text{QED}$$

Theorem 4.2. *Bessel's inequality.* If $X = \{x_1, \cdots, x_n\}$ is any finite orthonormal set in an inner product space, and x is any vector, then

$$||x||^2 \geq \sum_i |\alpha_i|^2, \tag{4.7}$$

where $\alpha_i = (x_i, x)$. Furthermore, the vector $x' \equiv x - \sum_i \alpha_i x_i$ is orthogonal to each x_j.

Proof. We first prove the inequality

$$0 \leq ||x'||^2 = (x', x') = \left(x - \sum_i \alpha_i x_i, x - \sum_j \alpha_j x_j\right)$$
$$= (x, x) - \sum_i \alpha_i^* (x_i, x) - \sum_j \alpha_j (x, x_j) + \sum_{i,j} \alpha_i^* \alpha_j (x_i, x_j)$$
$$= ||x||^2 - \sum_i |\alpha_i|^2 - \sum_j |\alpha_j|^2 + \sum_j |\alpha_j|^2 = ||x||^2 - \sum_i |\alpha_i|^2,$$

where we have used $(x_i, x_j) = \delta_{ij}$. Thus

$$||x||^2 \geq \sum_i |\alpha_i|^2.$$

To prove the second part of the theorem, we simply calculate

$$(x', x_j) = (x, x_j) - \sum_i \alpha_i^* (x_i, x_j) = \alpha_j^* - \alpha_j^* = 0. \quad \text{QED}$$

There are a number of ways to characterize a complete orthonormal set. Halmos has collected these and presented the proof of their equivalence in the form of a theorem.

Because it illustrates a very important logical technique, we adopt essentially his statement of the theorem and his logical route in its proof.*

Theorem 4.3. If $X = \{x_i\}$ is a finite orthonormal set of m vectors in an inner-product space V, the following six conditions on X are equivalent to each other:

1. X is complete;
2. if $(x_i, x) = 0$ for $i = 1, \cdots, m$, then $x = 0$;
3. X spans V,
4. if $x \in V$, then $x = \sum_{i=1}^m (x_i, x) x_i$,
5. if $x, y \in V$, then $(y, x) = \sum_{i=1}^m (y, x_i)(x_i, x)$—Parseval's equation;
6. if $x \in V$, then $||x||^2 = \sum_{i=1}^m |(x_i, x)|^2 = \sum_{i=1}^m |\alpha_i|^2$, in the notation of Theorem 4.2; i.e., the equal sign holds in Bessel's inequality.

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* See Halmos, P. R., *Finite-Dimensional Vector Spaces*, Second Edition. Princeton, N. J.: D. Van Nostrand, 1958. p. 124.

Proof. In this proof, we shall establish the implications

$$(1) \Longrightarrow (2) \Longrightarrow (3) \Longrightarrow (4) \Longrightarrow (5) \Longrightarrow (6) \Longrightarrow (1).$$

Thus, given any one of the six conditions, any of the other five can be reached by a logical chain, which establishes their logical equivalence. To begin, we assume that (1) is true and derive (2).

(1) $\Longrightarrow$ (2). We actually show this in the equivalent form: not (2) $\Longrightarrow$ not (1). If $(x_i, x) = 0$ for all i and $x \neq 0$, then we may adjoin $x/||x||$ to X and thus obtain an orthonormal set larger than X. Hence x is not complete according to Definition (4.4) of a complete set.

(2) $\Longrightarrow$ (3). Again, we actually show: not (3) $\Longrightarrow$ not (2). If X does not span V, then there is some $x \in V$ which cannot be written as a linear combination of the x_i, and therefore there exist no constants α_i such that

$$x = \sum_i \alpha_i x_i .$$

In particular, the set of constants $\alpha_i = (x_i, x)$ will be such that

$$x \neq \sum_i \alpha_i x_i .$$

Hence

$$x' = x - \sum_i (x_i, x) x_i \neq 0 ,$$

and yet, by the second part of Theorem 4.2, x' is orthogonal to each x_i. This statement is equivalent to "not (2)."

(3) $\Longrightarrow$ (4). If every x may be written in the form

$$x = \sum_j \alpha_j x_j ,$$

then

$$(x_i, x) = \sum_j \alpha_j (x_i, x_j) = \sum_j \alpha_j \delta_{ij} = \alpha_i ,$$

so that

$$x = \sum_i (x_i, x) x_i .$$

(4) $\Longrightarrow$ (5). If

$$x = \sum_i \alpha_i x_i \qquad \text{and} \qquad y = \sum_j \beta_j x_j ,$$

with $\alpha_i = (x_i, x)$ and $\beta_j = (x_j, y)$, then

$$\begin{aligned}(y, x) &= \Big(\sum_j \beta_j x_j, \sum_i \alpha_i x_i\Big) \\ &= \sum_{i,j} \beta_j^* \alpha_i (x_j, x_i) = \sum_i \beta_i^* \alpha_i = \sum_i (y, x_i)(x_i, x) .\end{aligned}$$

(5) $\Longrightarrow$ (6). Set $x = y$ in (5).

(6) $\Longrightarrow$ (1). [Actually, not (1) $\Longrightarrow$ not (6).] Assume that X is not complete. It is therefore contained in a larger orthogonal set, i.e., there exists a nonzero vector, x_0, orthogonal to each x_i in X; that is, $(x_i, x_0) = 0$, for all i. Thus

$$||x_0||^2 \neq \sum_i |(x_i, x_0)|^2 ,$$

since the right-hand side vanishes and the left does not. This statement is precisely "not (6)." This completes the proof of the equivalence of the six conditions.

Theorem 4.4. *Schwarz's inequality.* If x, y are vectors in an inner-product space, then

$$|(x, y)| \leq ||x|| \cdot ||y|| . \tag{4.8}$$

Proof. If $y = 0$, both sides vanish. If $y \neq 0$, then the set consisting of the single vector $y/||y||$ is orthonormal, and therefore by Bessel's inequality

$$\left|\left(x, \frac{y}{||y||}\right)\right|^2 \leq ||x||^2 \Longrightarrow |(x, y)| \leq ||x|| \cdot ||y|| . \quad \text{QED}$$

Examples of Schwarz's inequality.

1. In two-dimensional Euclidean space, the Schwarz inequality just says that $|\cos\theta| \leq 1$.

2. In n-dimensional unitary space, the Schwarz inequality is known as the Cauchy inequality. It asserts that for any two sequences of complex numbers, $(\alpha_1, \cdots, \alpha_n)$ and $(\beta_1, \cdots, \beta_n)$,

$$\left|\sum_{i=1}^{n} \alpha_i^* \beta_i\right|^2 \leq \sum_{i=1}^{n} |\alpha_i|^2 \cdot \sum_{i=1}^{n} |\beta_i|^2 . \tag{4.9}$$

3. In the space P, Schwarz's inequality becomes

$$\left|\int_0^1 x(t)^* y(t)\, dt\right|^2 \leq \int_0^1 |x(t)|^2\, dt \cdot \int_0^1 |y(t)|^2\, dt . \tag{4.10}$$

4.3 COMPLETE ORTHONORMAL SETS

It is very handy to have an orthonormal set of vectors as a basis for a vector space. Now we shall show by induction how one may construct an orthonormal basis from an arbitrary basis using a technique known as the *Gram-Schmidt orthogonalization process*. From Theorem 4.3 [(3) $\Longrightarrow$ (1)], this orthonormal basis will be a complete orthonormal set.

Let $X = \{x_1, \cdots, x_n\}$ be any basis in V. We shall construct a complete orthonormal set $Y = \{y_1, \cdots, y_n\}$ such that each y_j is a linear combination of

$x_1, \cdots, x_j$. Now $x_1 \neq 0$ because X is linearly independent; thus we may take $y_1 = x_1/||x_1||$. Now we put $y_2 = (x_2 - \alpha_1 y_1)/||x_2 - \alpha_1 y_1||$ and determine α_1 so that $(y_1, y_2) = 0$. Thus

$$(y_1, y_2) = \frac{(y_1, x_2) - \alpha_1}{||x_2 - \alpha_1 y_1||} = 0 \Longrightarrow \alpha_1 = (y_1, x_2) .$$

Therefore

$$y_2 = \frac{x_2 - (y_1, x_2) y_1}{||x_2 - (y_1, x_2) y_1||} .$$

Note that

$$x_2 - (y_1, x_2) y_1 = x_2 - \frac{(y_1, x_2)}{||x_1||} x_1 \neq 0 ,$$

since X is linearly independent. Thus we have found two vectors, y_1 and y_2, such that $(y_i, y_j) = \delta_{ij}$, $i, j = 1, 2$.

Now we discuss the induction step: Suppose that $y_1, \cdots, y_m$, $(m < n)$, have been found so that they form an orthonormal set and so that each y_j $(j = 1, \cdots, m)$ is a linear combination of $x_1, \cdots, x_j$. Let

$$y_{m+1} = \frac{x_{m+1} - (\alpha_1 y_1 + \cdots + \alpha_m y_m)}{||x_{m+1} - (\alpha_1 y_1 + \cdots + \alpha_m y_m)||} .$$

Then, for $j \leq m$,

$$(y_j, y_{m+1}) = \frac{(y_j, x_{m+1}) - \alpha_j}{||x_{m+1} - (\alpha_1 y_1 + \cdots + \alpha_m y_m)||} = 0 \Longrightarrow \alpha_j = (y_j, x_{m+1}) .$$

which implies that $\alpha_j = (y_j, x_{m+1})$. Thus

$$y_{m+1} = \frac{x_{m+1} - [(y_1, x_{m+1}) y_1 + \cdots + (y_m, x_{m+1}) y_m]}{||x_{m+1} - [(y_1, x_{m+1}) y_1 + \cdots + (y_m, x_{m+1}) y_m]||} . \tag{4.11}$$

The linear independence of the basis X ensures us that $y_{m+1} \neq 0$. When this process has been repeated n times, we shall have generated an orthonormal set of n basis vectors.

A matrix A is defined as a representation of the linear transformation A with respect to some basis $X = \{x_1, \cdots, x_n\}$. If we take the basis X to be an orthonormal set (or make it so by the Gram-Schmidt process), then we have a simple relationship between matrix elements and inner products which is used constantly in quantum mechanics. From Eq. 3.7, we have

$$Ax_j = \sum_k a_{kj} x_k .$$

If we now assume the x_i's to be an *orthonormal* basis and form the inner product (x_i, Ax_j), we have

$$(x_i, Ax_j) = \left(x_i, \sum_k a_{kj} x_k\right) = \sum_k a_{kj} (x_i, x_k) = \sum_k a_{kj} \delta_{ik} = a_{ij} .$$

Therefore the matrix elements of a linear transformation with respect to an orthonormal basis are given by

$$a_{ij} = (x_i, Ax_j) \ . \tag{4.12}$$

All that is being done in the Gram-Schmidt process is to start with some vector x_1, normalize it, and then subtract from the next vector, x_2, any component of the normalized x_1 which it may contain. After normalizing the modified x_2, one goes on to x_3 and removes all components of the previous normalized vectors. The process can be continued indefinitely, although clearly in a finite-dimensional space it must eventually terminate. At each step of the operation, one *decreases* (or leaves unchanged) the norm of the vector in question, since one subtracts some of its components. This fact is the basis for a very simple application of the Gram-Schmidt process to the theory of determinants. Let us denote a matrix A by $\{a_1, a_2, \cdots, a_n\}$, where a_1 is the first row, a_2 is the second row, etc. Now if the row vectors a_i form an orthogonal set of vectors, clearly det $(AA^\dagger) = ||a_1||^2||a_2||^2 \cdots ||a_n||^2$. Since, according to Eq. (3.17) det $A^\dagger = (\det A)^*$, it follows that

$$|\det A| = ||a_1||\ ||a_2||\ \cdots\ ||a_n|| \ .$$

Here the norm of the row vector a_i has its usual meaning, that is,

$$||a_i|| = \left(\sum_{j=1}^{n} a_{ij}^* a_{ij}\right)^{1/2}$$

[see Eq. (4.5)].

Now consider an arbitrary matrix $B = \{b_1, b_2, \cdots, b_n\}$. We know that to any row of B one can add an arbitrary multiple of any other row and leave its determinant unchanged. Thus, by the Gram-Schmidt process, we can create from the row vectors of B a new matrix C which we denote by

$$\{\beta_1 c_1, \beta_2 c_2, \cdots \beta_n c_n\} \ ,$$

where $\{c_1, c_2, \cdots, c_n\}$ is a set of orthonormal vectors and $\{\beta_1, \beta_2, \cdots, \beta_n\}$ is a set of scalars, each of which is respectively less than or equal to $||b_1||$, $||b_2||$, $\cdots$, $||b_n||$. This follows from Eq. (4.11). Now since we have added multiples of rows to other rows, det C = det B. Moreover, the c_i are normalized to 1, so we find (just as for det A, above) that

$$|\det B| = |\det C| = |\beta_1|\ |\beta_2|\ \cdots\ |\beta_n| \ .$$

Since $|\beta_i| \leq ||b_i||$, we have

Hadamard's inequality. If b_i denotes the ith row of the matrix B, then

$$|\det B| \leq ||b_1||\ ||b_2||\ \cdots\ ||b_n|| \ .$$

This result is useful in the theory of integral equations.

We close this section with a simple but important result that will be required continually in the proofs of subsequent theorems.

Theorem 4.5. A linear transformation A on an inner-product space is the zero transformation if and only if $(x, Ay) = 0$, for all x and y.

Proof. If $A = 0$, then $(x, Ay) = (x, 0) = 0$ (by property 2 in the definition of the inner product). Conversely, if $(x, Ay) = 0$ for all x and y, we take $x = Ay$. Then $(Ay, Ay) = 0 \Longrightarrow Ay = 0$, for all y (by positive definiteness of the inner product) and therefore $A = 0$. QED

4.4 SELF-ADJOINT (HERMITIAN AND SYMMETRIC) TRANSFORMATIONS

The most important linear transformations in quantum physics are the so-called *self-adjoint operators*, which will be defined presently. First, however, we must define the adjoint transformation.

Definition 4.5. Let A be a linear transformation defined on a vector space V. For every A, the operator $A^\dagger$, which satisfies the equation

$$(Ax, y) = (x, A^\dagger y) , \qquad \text{for all } x, y \tag{4.13}$$

is called the *adjoint* of A.

Since

$$(x, Ay) = (Ay, x)^* = (y, A^\dagger x)^* = (A^\dagger x, y) ,$$

the adjoint may be defined equivalently by

$$(x, Ay) = (A^\dagger x, y) , \qquad \text{for all } x, y . \tag{4.14}$$

To be strictly rigorous, we ought at this point to prove that the adjoint of a given linear transformation always exists. We leave the proof to the reader (Problem 18), however, since it is of little interest for the purposes we have in mind. The interested reader may find a coordinate-independent proof in a text on vector spaces.*

Moreover, the adjoint $A^\dagger$ is unique. Suppose that

$$(x, By) = (Ax, y) = (x, Cy)$$

for all x, y. Then $(x, [B - C]y) = 0$ for all x, y which implies $B = C$ by Theorem 4.5.

To show that $A^\dagger$ is in fact a *linear* transformation, we compute

$$\begin{aligned}(x, A^\dagger[\alpha y + \beta z]) &= (Ax, [\alpha y + \beta z]) = (Ax, \alpha y) + (Ax, \beta z) \\ &= \alpha(Ax, y) + \beta(Ax, z) = \alpha(x, A^\dagger y) + \beta(x, A^\dagger z) \\ &= (x, \alpha A^\dagger y) + (x, \beta A^\dagger z) .\end{aligned}$$

Therefore

$$(x, A^\dagger[\alpha y + \beta z] - \alpha A^\dagger y - \beta A^\dagger z) = 0 \Longrightarrow A^\dagger(\alpha y + \beta z) = \alpha A^\dagger y + \beta A^\dagger z ,$$

so $A^\dagger$ is a linear operator. We now prove a few simple properties of the adjoint.

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* For example, Hoffman and Kunze, p. 237.

Theorem 4.6.

1. $(A + B)^\dagger = A^\dagger + B^\dagger$.
2. $(AB)^\dagger = B^\dagger A^\dagger$.
3. $(\alpha A)^\dagger = \alpha^* A^\dagger$, where α is a scalar.
4. $(A^\dagger)^\dagger = A$.

Proof.

1. For all x and y,

$$\begin{aligned}(x, [A + B]^\dagger y) &= ([A + B]x, y) \\ &= (Ax + Bx, y) = (Ax, y) + (Bx, y) \\ &= (x, A^\dagger y) + (x, B^\dagger y) \\ &= (x, [A^\dagger + B^\dagger]y) .\end{aligned}$$

The result now follows from Theorem 4.5.

2. $(x, [AB]^\dagger y) = (ABx, y) = (Bx, A^\dagger y) = (x, B^\dagger A^\dagger y)$, and again, using Theorem 4.5, the result follows. QED

We leave the proofs of properties (3) and (4), which are very similar to those we have done, to the reader.

Theorem 4.7. Let the matrix of A be (a_{ij}) with respect to an orthonormal basis X. Then the matrix of $A^\dagger$ with respect to X is $[A^\dagger]_{ij} = a^*_{ji}$, that is, $[A^\dagger] = [\tilde{A}]^*$.

Proof. From Eq. (4.12), we have

$$a_{ij} = (x_i, Ax_j) .$$

Therefore

$$a_{ij} = (A^\dagger x_i, x_j) = (x_j, A^\dagger x_i)^* = [A^\dagger]^*_{ji} \Longrightarrow [A^\dagger]_{ij} = a^*_{ji} . \quad \text{QED}$$

Thus the definition of the adjoint of an operator reduces to that of the adjoint of a matrix, given in Section 3.7.

Definition 4.6. If $A = A^\dagger$, A is its own adjoint, and A is called a *self-adjoint* operator. In real inner-product spaces, a self-adjoint operator is called a *symmetric* operator, and in a complex inner-product space, a *Hermitian* operator.

The operators that correspond to physical observables in quantum mechanics are all Hermitian operators.

It follows immediately from Theorem 4.6 that if A and B are self-adjoint, then so is their sum. Similarly, if A is self-adjoint, then the operator αA will be self-adjoint if and only if α is real. We also derive the following simple result:

Theorem 4.8. If A and B are self-adjoint, then AB is self-adjoint if and only if A and B commute (i.e., if and only if their commutator, $[A, B]$, vanishes).

Proof. If $AB = BA$, then

$$(AB)^\dagger = B^\dagger A^\dagger = BA = AB ,$$

so AB is self-adjoint. If $(AB)^\dagger = AB$, then

$$AB = (AB)^\dagger = B^\dagger A^\dagger = BA , \qquad \text{so } [A, B] = 0 . \quad \text{QED}$$

The reason why a self-adjoint operator on a real inner-product space is called a symmetric operator is apparent from Theorem 4.7. For if $A = A^\dagger$, the matrix representations satisfy

$$a_{ij} = [A^\dagger]_{ij} = a_{ji}^* = a_{ji} ,$$

because the matrix elements belong to the field of real numbers. Thus $A = \tilde{A}$, and A is what we have previously called a symmetric matrix.

We now prove two theorems that are variations of Theorem 4.5 and will play a role much like the role which that theorem has played in many subsequent proofs. Basically, we are concerned with strengthening that theorem for more restricted applications to self-adjoint operators.

Theorem 4.9. Let A be a self-adjoint operator in a real inner-product space. Then $A = 0$ if and only if $(x, Ax) = 0$, for all x.

Proof. If $A = 0$, then $(x, Ax) = 0$. Thus the proof of the necessity of the condition is trivial. However, its sufficiency is not. To establish sufficiency consider the expression

$$(x + y, A[x + y]) = (x, Ax) + (y, Ay) + (x, Ay) + (y, Ax) .$$

From this we get

$$(x, Ay) + (y, Ax) = (x + y, A[x + y]) - (x, Ax) - (y, Ay)$$

for all x and y. Since each term on the right-hand side is of the form (x, Ax), which is zero by hypothesis, it follows that

$$(x, Ay) + (y, Ax) = 0 .$$

But

$$(y, Ax) = (A^\dagger y, x) = (Ay, x) = (x, Ay)^* = (x, Ay) ,$$

since $A^\dagger = A$ and the inner product is a real number. Therefore $(x, Ay) = 0$ for all x and y, which, by Theorem 4.5, implies that $A = 0$. QED

If we know that the inner-product space is complex, as is the case in most of the applications to physics, we may strengthen this result still further.

Theorem 4.10. If A is any linear transformation on a unitary space, then $A = 0$ if and only if $(x, Ax) = 0$ for all x.

Proof. As in the previous theorem, necessity is obvious. The proof of sufficiency follows easily from a result we obtained in proving the last theorem. We showed that

$$(x, Ay) + (y, Ax) = 0$$

for all x and y, if $(x, Ax) = 0$ for all x. No further conditions on A were used in establishing this result, so we are free to use it here. Since it holds for all y, we may replace y by iy to obtain

$$(x, Aiy) + (iy, Ax) = i[(x, Ay) - (y, Ax)] = 0 .$$

It follows that

$$(x, Ay) - (y, Ax) = 0 ,$$

for all x and y. Adding this to the result above, we have $(x, Ay) = 0$ for all x and y. Theorem 4.5 now implies that $A = 0$. QED

As a consequence of Theorems 4.9 and 4.10, we have the following result.

Corollary. If A is a self-adjoint linear transformation in an inner-product space (real *or* complex), then $A = 0$ if and only if $(x, Ax) = 0$, for all x.

A fundamental result for quantum mechanics is the following:

Theorem 4.11. A linear transformation A, on a unitary space, is Hermitian if and only if (x, Ax) is real for all x.

Proof. If $A = A^\dagger$, then $(x, Ax) = (A^\dagger x, x) = (x, Ax)^*$, so (x, Ax) is real. Conversely, if (x, Ax) is always real, then

$$(x, Ax) = (x, Ax)^* = (Ax, x) = (x, A^\dagger x) .$$

Therefore $(x, [A - A^\dagger]x) = 0$ for all x, and from Theorem 4.10 it follows that $A = A^\dagger$. QED

Note that this theorem is not true in a real inner-product space; i.e., if (x, Ax) is real it does not follow that A is symmetric. Clearly, (x, Ax) is always real in a real inner-product space; but in a complex inner-product space its reality for all x has the important consequence that A is Hermitian.

The quantity (x, Ax) is called the *expectation value* of the operator A in quantum mechanics. For normalized wave functions, it is the average value of many measurements of the observable A on systems in the state described by x. Naturally, this number must be real. Therefore the operators of quantum mechanics corresponding to physical observable must be Hermitian.

A closely related result is that the eigenvalues of a Hermitian matrix are all real numbers. Since in quantum theory the only possible results of a physical measurement are the eigenvalues of the corresponding Hermitian operator, this result, which we prove next, is also fundamental to the theory.

Theorem 4.12. The eigenvalues of a Hermitian operator are real.

Proof.

$$Ax = \lambda x \Longrightarrow (x, Ax) = \lambda ||x||^2 \Longrightarrow \lambda = \frac{(x, Ax)}{||x||^2} ,$$

which, by Theorem 4.11, is real if A is Hermitian. QED

Example. The Hamiltonian for the simple harmonic oscillator is

$$H = (p^2/2m) + \tfrac{1}{2} m\omega^2 x^2 .$$

Earlier [Example 2, Section 3.10] we claimed that its eigenvalues were all positive. We can now give a simple demonstration of this fact. Let $Hu = Eu$, where the eigenvectors u are normalized, that is, $||u|| = 1$. Then

$$\begin{aligned} E = (u, Hu) &= \left(u, \frac{p^2}{2m} u\right) + (u, \tfrac{1}{2} m\omega^2 x^2 u) \\ &= \frac{1}{2m} (p^\dagger u, pu) + \tfrac{1}{2} m\omega^2 (x^\dagger u, xu) \\ &= \frac{1}{2m} (pu, pu) + \tfrac{1}{2} m\omega^2 (xu, xu) , \end{aligned}$$

since p and x are both Hermitian. Thus

$$E = (1/2m)||pu||^2 + \tfrac{1}{2} m\omega^2 ||xu||^2 \geq 0 ,$$

since $m, \omega > 0$.

It is clear from this example that the eigenvalues of the square of *any* Hermitian operator will always be positive or zero. This also follows from Theorem 4.12.

Example. In our earlier treatment of the eigenvalue-eigenvector for the harmonic-oscillator problem, we stated that the nth normalized eigenvector was

$$u_n = c_n (a^\dagger)^n u_0 , \tag{4.15}$$

where $c_n = (1/n!)^{1/2}$ [Eqs. (3.48) and (3.49)]. We can now provide a simple proof of this fact by the same operator algebra techniques.

The creation and annihilation operators

$$a^\dagger = (xm\omega - ip)(2m\omega\hbar)^{-1/2} , \qquad a = (xm\omega + ip)(2m\omega\hbar)^{-1/2}$$

are adjoints of each other, as the notation has anticipated; p and x are themselves Hermitian.

Normalization of the eigenvectors u_n requires that $(u_n, u_n) = 1$. Since

$$u_n = \frac{c_n}{c_{n-1}} c_{n-1} a^\dagger (a^\dagger)^{n-1} u_0 = \frac{c_n}{c_{n-1}} a^\dagger u_{n-1} ,$$

we may write this as

$$\begin{aligned} 1 &= \left(\frac{c_n}{c_{n-1}} a^\dagger u_{n-1}, \frac{c_n}{c_{n-1}} a^\dagger u_{n-1}\right) = \left|\frac{c_n}{c_{n-1}}\right|^2 (a^\dagger u_{n-1}, a^\dagger u_{n-1}) \\ &= \left|\frac{c_n}{c_{n-1}}\right|^2 (u_{n-1}, aa^\dagger u_{n-1}) = \left|\frac{c_n}{c_{n-1}}\right|^2 (\hbar\omega)^{-1} \left(u_{n-1}, \left(H + \frac{\hbar\omega}{2}\right) u_{n-1}\right) , \end{aligned}$$

where we have used Eq. (3.44). Now using Eq. (3.48), we obtain

$$1 = \left|\frac{c_n}{c_{n-1}}\right|^2 n(u_{n-1}, u_{n-1}) = \left|\frac{c_n}{c_{n-1}}\right|^2 n .$$

Therefore

$$|c_n|^2 = \frac{1}{n}|c_{n-1}|^2 .$$

By iterating this, we obtain

$$|c_n|^2 = \frac{1}{n}\frac{1}{n-1}|c_{n-2}|^2 = \frac{1}{n}\frac{1}{n-1}\frac{1}{n-2}|c_{n-3}|^2 = \cdots = \frac{1}{n!}|c_0|^2 = \frac{1}{n!} ,$$

since $c_0 = 1$. Therefore, the choice

$$c_n = \left(\frac{1}{n!}\right)^{1/2}$$

will ensure that all the eigenfunctions generated by (4.15) are normalized.

4.5 ISOMETRIES—UNITARY AND ORTHOGONAL TRANSFORMATIONS

We now introduce another very important class of linear transformations.

Definition 4.7. If a linear transformation U on an inner-product space is such that $U^\dagger U = I$, U is called an isometry (or *isometric* transformation). If, in addition, $UU^\dagger = I$, then the isometry U is called a *unitary* transformation. A linear transformation A satisfying $\tilde{A}A = I$ is called an *orthogonal* transformation.

In finite-dimensional spaces, the existence of a left inverse, $U^\dagger$ ($U^\dagger U = I$), implies that $U^\dagger$ is also a right inverse ($UU^\dagger = I$). Thus our definition need make no mention of left and right in finite-dimensional spaces. It follows that if U is an isometry in a finite-dimensional space, then U is invertible and hence unitary. Moreover, in this case we have:

$$(U^{-1})^\dagger = (U^\dagger)^\dagger = U = (U^{-1})^{-1} .$$

We discuss several alternative characterizations of isometric transformations in the following theorem, which is applicable to any vector space, whether finite- or infinite-dimensional.

Theorem 4.13. The following three conditions on a linear transformation U on an inner-product space are equivalent to each other.

1. $U^\dagger U = I$;
2. $(Ux, Uy) = (x, y)$ for all x and y;
3. $||Ux|| = ||x||$ for all x.

Proof. If $U^\dagger U = I$, then $(Ux, Uy) = (x, U^\dagger Uy) = (x, y)$ for all x and y, and, in particular, if $y = x$, $||Ux||^2 = ||x||^2$ for all x. This proves that

$$(1) \Longrightarrow (2) \Longrightarrow (3) .$$

To show (3) $\Longrightarrow$ (1) and thus establish the equivalence of the three conditions, we note that

$$||Ux||^2 = (Ux, Ux) = (U^\dagger Ux, x) = (x, x) \Longrightarrow ([U^\dagger U - I]x, x) = 0 ,$$

for all x. Now, if the operator $[U^\dagger U - I]$ is self-adjoint, we may use the corollary to Theorem 4.10 to deduce $U^\dagger U = I$, which would complete the proof. Since

$$[U^\dagger U - I]^\dagger = (U^\dagger U)^\dagger - I^\dagger = U^\dagger (U^\dagger)^\dagger - I = [U^\dagger U - I] ,$$

$[U^\dagger U - I]$ is self-adjoint. QED

This theorem explains why an isometry is so called. Property 3 tells us that these transformations preserve lengths. Recalling that the cosine of angle θ between the vectors x and y is given by

$$\cos\theta = \frac{(x, y)}{||x||\,||y||} ,$$

we see that isometric linear transformations also preserve angles between vectors because they preserve scalar products (property 2) and lengths (property 3). This holds in n-dimensional space, too, where the definition of the "angle" between vectors is formally identical to this one.

Theorem 4.14. If $\{x_i\}$ is a complete orthonormal set, then so is $\{Ux_i\}$ for any isometry U.

Proof.

$$(Ux_i, Ux_j) = (x_i, U^\dagger Ux_j) = (x_i, x_j) = \delta_{ij} ,$$

so the set $\{Ux_i\}$ is orthonormal.

To prove the completeness of $\{Ux_i\}$, we show that the set $\{Ux_i\}$ satisfies Parseval's identity (condition 5 of Theorem 4.3). We have

$$\sum_i (y, Ux_i)(Ux_i, x) = \sum_i (U^\dagger y, x_i)(x_i, U^\dagger x) = (U^\dagger y, U^\dagger x) ,$$

since the set $\{x_i\}$ is complete. But

$$(U^\dagger y, U^\dagger x) = (y, UU^\dagger x) = (y, x) .$$

Therefore, Parseval's identity is satisfied, and the set $\{Ux_i\}$ is complete. QED

Isometries, therefore, are linear transformations that convert one orthonormal basis into another. We have previously seen an example of such transformations in Section 1.4, where we considered rotations of basis vectors in three-dimensional Cartesian space. There we defined a vector as a triple of numbers, x_i, which upon rotation of the basis vectors transform according to the rule

$$x_j' = \sum_i a_{ij} x_i ,$$

where the coefficients a_{ij} satisfy the relations

$$\sum_k a_{ik}a_{jk} = \sum_k a_{ki}a_{kj} = \delta_{ij} .$$

[See Eqs. (1.13).] Now let $[U]_{ij} = u_{ij}$, so that $[U^\dagger]_{ij} = u_{ji}^*$. The matrix representation of the operator equation, $UU^\dagger = U^\dagger U = I$, is

$$[UU^\dagger]_{ij} = \sum_k u_{ik}u_{jk}^* = [U^\dagger U]_{ij} = \sum_k u_{ki}^* u_{kj} = \delta_{ij} . \tag{4.16}$$

In a real inner-product space, $u_{ij} = u_{ij}^*$, and Eq. (4.16) reduces to the orthogonality relations above. We anticipated this result when we called rotations of vectors in three dimensions *orthogonal transformations.*

It was pointed out that the determinants of the various rotation matrices of Section 1.4 all had the value 1. We now prove the related general result.

Theorem 4.15. If A is an orthogonal matrix, then $\det A = \pm 1$.

Proof. If A is an orthogonal matrix then by definition $\tilde{A}A = I$. Thus

$$\det(\tilde{A}A) = \det \tilde{A} \det A = (\det A)^2 = 1 \Longrightarrow \det A = \pm 1 ,$$

where we have used Eq. (3.17). QED

The $+1$ corresponds to a rotation, and the -1 corresponds to a rotation plus an *inversion*—that is, a switch from a right-handed to a left-handed coordinate system. If U is a unitary matrix, then a similar proof shows that $|\det U| = 1$, or $\det U = e^{i\phi}$, where ϕ is any real number (Problem 1).

4.6 THE EIGENVALUES AND EIGENVECTORS OF SELF-ADJOINT AND ISOMETRIC TRANSFORMATIONS

We are now in a position to prove some general theorems about the spectra of certain linear operators. These theorems are of fundamental importance in quantum mechanics, where we are concerned with finding the eigenvalues and eigenvectors of Hermitian operators.

Theorem 4.16. If A is a self-adjoint linear transformation, then every eigenvalue of A is real.

Proof. The eigenvalues of A are the λ such that $Ax = \lambda x$, where $x \neq 0$. Now $(x, Ax) = (x, \lambda x) = \lambda(x, x)$, and since A is self-adjoint,

$$(x, Ax) = (Ax, x) = (\lambda x, x) = \lambda^*(x, x).$$

Since x is an eigenvector, $x \neq 0$, and hence $(x, x) \neq 0$. Therefore $\lambda = \lambda^*$, so λ is real. QED

This result is only a trivial extension of Theorem 4.12. We include it both to demonstrate this alternative proof—which is the one usually given in the context of quantum mechanics—and to collect together in one section all the

main results on the spectra of the operators we have studied. We turn now to the spectra of isometries.

Theorem 4.17. Every eigenvalue of an isometry has absolute value 1.

Proof. If U is an isometry and $Ux = \lambda x$ $(x \neq 0)$ then

$$||x|| = ||Ux|| = |\lambda|\ ||x|| \Longrightarrow |\lambda| = 1$$

since $||x|| \neq 0$. QED

In the proof of this theorem, we know that $||x|| = (x, x)^{1/2} \neq 0$ because $x \neq 0$, and our inner product is such that $(x, x) = 0$ if and only if $x = 0$.

We now prove the central theorem regarding the spectral properties of the transformations we have been considering.

Theorem 4.18. If A is either self-adjoint or isometric, then eigenvectors of A belonging to *distinct* eigenvalues are orthogonal.

Proof. Suppose that $Ax_1 = \lambda_1 x_1$ and $Ax_2 = \lambda_2 x_2$, $\lambda_1 \neq \lambda_2$.

a) If A is self-adjoint, then

$$(x_1, Ax_2) = \lambda_2(x_1, x_2)$$

and also

$$(x_1, Ax_2) = (Ax_1, x_2) = \lambda_1^*(x_1, x_2) = \lambda_1(x_1, x_2)\ ,$$

since $A = A^\dagger$ and therefore the eigenvalues are real (Theorem 4.16). Therefore, $(\lambda_1 - \lambda_2)(x_1, x_2) = 0$, and hence $(x_1, x_2) = 0$, since $\lambda_1 \neq \lambda_2$.

b) If A is an isometry, then

$$(x_1, x_2) = (Ax_1, Ax_2) = \lambda_1^*\lambda_2(x_1, x_2) = (\lambda_2/\lambda_1)(x_1, x_2)\ ,$$

since by Theorem 4.17 $|\lambda|^2 = \lambda\lambda^* = 1$ for an isometry. Therefore, since $\lambda_1 \neq 0$ we have

$$(\lambda_1 - \lambda_2)(x_1, x_2) = 0\ .$$

But $\lambda_1 \neq \lambda_2$, so we get $(x_1, x_2) = 0$. QED

When we solve the characteristic polynomial for an $n \times n$ self-adjoint or isometric transformation we get n eigenvalues. If any two are different, the corresponding eigenvectors are orthogonal, and if they are all distinct, then the eigenvectors are all orthogonal and span the space on which the matrix is defined. If k, $(k \leq n)$ of them are the same $(= \lambda_0)$, it can and will be shown (Theorem 4.20) that there always exist k linearly independent (but not necessarily orthogonal) eigenvectors belonging to λ_0. Of course all these k eigenvectors are orthogonal to all the other $(n - k)$ which belong to eigenvalues unequal to λ_0. And these k linearly independent eigenvectors belonging to λ_0 may be replaced by an orthogonal set of k eigenvectors by applying the Gram-Schmidt orthogonalization process to them. This procedure is often followed in quantum mechanics for degenerate eigenvalues. We can summarize by saying that the *algebraic* multiplicity (the number of times a given root is repeated

in the solution to the characteristic equation) equals the *geometric* multiplicity (the number of linearly independent eigenvectors corresponding to that root) for self-adjoint and isometric linear transformations. Therefore *the eigenvectors of self-adjoint and isometric linear transformations span the space and can be chosen to be a complete orthonormal set.* We shall prove this fundamental result in the next section.

Now we shall look at an example of the use of the preceding theorems.

Example. *The rotation matrix and the Lorentz matrix.* We want to examine the rotation matrix and the Lorentz matrix from the point of view of the previous theorems on eigenvalues and eigenvectors. The two-dimensional rotation matrix, which we denote here by R, is given by

$$R = \begin{bmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{bmatrix}.$$

As we saw in Section 1.4, it describes the effect of rotating the basis of a vector space. The Lorentz matrix describes the effect of a Lorentz transformation. The reader is no doubt familiar with the equations of the Lorentz transformation which relate the space-time coordinates of two observers moving relative to each other with uniform velocity v along their common x_3x_3'-axis. For this configuration we have

$$\begin{aligned} x_1' &= x_1, \qquad & x_2' &= x_2 \\ x_3' &= \gamma(x_3 - vt), \qquad & t' &= \gamma(t - vx_3/c^2)\,, \end{aligned} \tag{4.17}$$

where $\gamma \equiv (1 - \beta^2)^{-1/2}$ and $\beta \equiv v/c$.

The transformation described by Eqs. (4.17) can also be expressed in matrix form. One device for doing so is to write $x_4 = ict$. This is motivated by the fact that then

$$\sum_{i=1}^{4} x_i^2 = x_1^2 + x_2^2 + x_3^2 - c^2t^2\,, \tag{4.18}$$

and this quantity is an invariant in special relativity expressing the invariance of the speed of light in different frames of reference. The coordinates $(x_1, x_2, x_3, x_4 = ict)$ are referred to as the coordinates of Minkowski space. If we now define a Lorentz transformation by

$$u' = Lu\,,$$

where $u = (u_1, u_2, u_3, u_4)$ is any four vector in Minkowski space and L is the Lorentz matrix,

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \gamma & i\beta\gamma \\ 0 & 0 & -i\beta\gamma & \gamma \end{bmatrix}, \tag{4.19}$$

then the reader can easily check that Lx gives precisely the transformed vector x' whose components are given by Eqs. (4.17).

A simple and useful parametric form for the Lorentz matrix results if we let

$$\gamma \equiv \frac{1}{\sqrt{1-\beta^2}} \equiv \cosh \alpha \,. \tag{4.20}$$

Both γ and $\cosh \alpha$ range from a minimum of $1 (\beta = 0)$ to a maximum of ∞ $(v = c,\ \beta = 1)$. It follows that

$$\sinh \alpha = (\cosh^2 \alpha - 1)^{1/2} = \frac{\beta}{\sqrt{1-\beta^2}} \,.$$

and

$$\alpha = \tanh^{-1} \beta \,.$$

The matrix L becomes

$$L = L(\alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cosh \alpha & i \sinh \alpha \\ 0 & 0 & -i \sinh \alpha & \cosh \alpha \end{bmatrix} . \tag{4.21}$$

Now let us examine the general properties of R and L. We first note that $\tilde{R}R = I$ and $\tilde{L}L = I$, that is, the inverse of each matrix is equal to its transpose. For the rotation matrix, this expresses the fact that the inverse of a rotation R through an angle ϕ is obtained by replacing ϕ by $-\phi$ in R, which is equivalent to transposing R. In the case of a Lorentz transformation, the inverse is seen on physical grounds to correspond to replacing v by $-v$ (that is, α by $-\alpha$) in the matrix, which is equivalent to transposing L. We may summarize by saying that both R and L are *orthogonal* matrices. However, virtually all the theorems proved above refer to unitary matrices, so if we want to draw any general conclusions about R and L, we had best see if R and L have the more powerful property of unitarity. Since the elements of R are real, $\tilde{R} = R^\dagger$, so we have immediately $R^\dagger R = I$, that is, R is indeed a unitary matrix. However, $\tilde{L} \neq L^\dagger$, because the elements of L are complex. A straightfoward multiplication shows that $L^\dagger L \neq I$, so L is *not* a unitary matrix. Thus our theorems on unitary transformations tell us nothing about L. They do, however, tell us that as far as rotations are concerned, R must have eigenvalues whose modulus is unity and eigenvectors which are orthogonal to each other (if the eigenvalues are distinct). A simple calculation shows that the two eigenvalues of R are given by $\lambda_+ = e^{i\phi}$ and $\lambda_- = e^{-i\phi}$. The corresponding eigenvectors are $x_+ = (1, i)$ and $x_- = (1, -i)$. Thus

$$|\lambda_\pm| = |e^{\pm i\phi}| = 1 \,,$$

which fulfills the prediction of Theorem 4.17, and

$$(x_+, x_-) = 0 \,,$$

which is in accordance with Theorem 4.18.

If we want to obtain similar information about L, we will have to rely on theorems which do not depend on unitarity. In this case we do not have far to look, since it is clear that $L^\dagger = L$, that is, L is a self-adjoint matrix. Thus we conclude immediately that the eigenvalues of L are real (Theorem 4.12) and that the eigenvectors corresponding to distinct eigenvalues are orthogonal (Theorem 4.18). A simple calculation (Problem 16) shows that the eigenvalues of L are $\lambda_1 = 1$, $\lambda_2 = 1$, $\lambda_+ = \cosh\alpha + \sinh\alpha$ and $\lambda_- = \cosh\alpha - \sinh\alpha$. They are real as expected from Theorem 4.12. The corresponding eigenvectors are $x_1 = (1, 0, 0, 0)$, $x_2 = (0, 1, 0, 0)$, $x_+ = (0, 0, 1, -i)$ and $x_- = (0, 0, 1, i)$. They are orthogonal in accordance with Theorem 4.18. Thus, only by considering L as a Hermitian matrix can we obtain any information about the spectral properties of L. The *orthogonality* of L has so far told us nothing. The only useful fact that we can derive from orthogonality is $\det L = \pm 1$ (Theorem 4.15). This result is also true for R (that is, $\det R = \pm 1$), since R is orthogonal as well as unitary. Note that for both the rotation matrix and the Lorentz matrix the determinant is equal to the product of the eigenvalues as we should expect from the invariance of the determinant under similarity transformations (Theorem 3.13). The reader can easily check that in this case the trace is also invariant under similarity transformations (Theorem 3.14).

Since one is perhaps inclined to think that there should be little difference between orthogonal matrices and unitary matrices, it may be disturbing to some readers that we have been unable to deduce any important consequences from the *orthogonality* of L. One might imagine that the difficulties with complex orthogonal matrices such as L come from the form of the inner product (with its unsymmetric complex conjugation) which we use, and that perhaps if we used a simple Euclidean type inner product such as

$$[x, y] = \sum_{i=1}^{n} \xi_i \eta_i \tag{4.22}$$

—which after all motivated the introduction to *real* orthogonal matrices in the first place (Chapter 1)—then we could save some theorems. Because of the missing complex conjugation, we are giving up several properties of the usual inner product, in particular, positive-definiteness. However, the *pseudo inner product* defined by Eq. (4.22) has certain attractive properties. It is linear:

$$[x + y, z] = [x, z] + [y, z]\,; \tag{4.23}$$

it is symmetric:

$$[x, y] = [y, x]\,; \tag{4.24}$$

and it is homogeneous:

$$[\alpha x, y] = \alpha[x, y]\,. \tag{4.25}$$

Note that in a *real* vector space, these properties are the same as those given by axioms (1) and (2) of Definition 4.1. Using these general properties, the reader

can investigate the subject of *pseudo inner-product spaces* along the lines which we have followed for inner-product spaces. Many of the results in this section and in the previous ones have close analogues in a pseudo inner-product space.

The main geometric problem is to elucidate the role of the *null-vector*, that is, a vector $x \neq 0$ for which $[x, x] = 0$. As far as operator theory goes, the key definition is that of the *transpose operator*, $\tilde{A}$, which satisfies $[x, Ay] = [\tilde{A}x, y]$ for all x and y. When represented by a matrix, A has just the form one should expect, and the study of *symmetric matrices* ($\tilde{S} = S$) and orthogonal matrices ($\tilde{O}O = I$) proceeds in much the same manner as in this section, except that certain difficulties associated with the existence of null-vectors must be dealt with. Among the results which one obtains is the fact that eigenvectors of a symmetric matrix belonging to distinct eigenvalues are pseudo orthogonal, i.e., if x_1 and x_2 are the eigenvectors in question, $[x_1, x_2] = 0$. Note that the eigenvalues of a symmetric operator need not be real in a general (complex) vector space. Also, if O is an orthogonal matrix (such as the Lorentz matrix), then every eigenvalue of O belonging to a *nonnull* eigenvector is equal to 1 or -1. The demonstration of these results and others we leave as a problem (Problem 4.27). It will be found that the theorems which one can obtain for orthogonal operators in a pseudo inner-product space are far weaker than the theorems for unitary operators in a true inner-product space.

We close this section on spectral theory with an interesting generalization that contains many of the previous theorems as special cases. It shows that these results really depended only on the fact that the linear transformations in question commuted with their adjoints.

Definition 4.8. A linear transformation is *normal* if and only if $AA^\dagger = A^\dagger A$.

All self-adjoint (Hermitian and symmetric) and isometric (unitary and real orthogonal) linear transformations are normal. In addition, anti-self-adjoint operators [anti-symmetric ($A = -\tilde{A}$) and anti-Hermitian ($A = -A^\dagger$) operators] are normal.

Theorem 4.19. If A is normal, then eigenvectors belonging to distinct eigenvalues are orthogonal.

Proof. We first prove the preliminary result, that if A is normal, then

$$Ax = \lambda x \Longleftrightarrow A^\dagger x = \lambda^* x .$$

Normality of A implies that

$$||Ax||^2 \equiv (Ax, Ax) = (A^\dagger Ax, x) = (AA^\dagger x, x) = (A^\dagger x, A^\dagger x) = ||A^\dagger x||^2 .$$

Now, if A is normal, $(A - \lambda)$ is normal, and since $(A - \lambda)^\dagger = A^\dagger - \lambda^*$, using $(A - \lambda)$ in place of A in the above, we obtain

$$||Ax - \lambda x||^2 = ||A^\dagger x - \lambda^* x||^2 .$$

It follows immediately that

$$Ax = \lambda x \Longleftrightarrow A^\dagger x = \lambda^* x .$$

Now we may prove the theorem itself. If $Ax_1 = \lambda_1 x_1$ and $Ax_2 = \lambda_2 x_2$, then

$$(x_1, Ax_2) = \lambda_2(x_1, x_2) ,$$

and

$$(x_1, Ax_2) = (A^\dagger x_1, x_2) = (\lambda_1^* x_1, x_2) = \lambda_1(x_1, x_2) .$$

Therefore

$$(\lambda_1 - \lambda_2)(x_1, x_2) = 0 \Longrightarrow (x_1, x_2) = 0 ,$$

if $\lambda_1 \neq \lambda_2$. QED

Any matrix A may be decomposed into the sum of a self-adjoint and an anti-self-adjoint matrix, for we have the identity

$$A \equiv \tfrac{1}{2}(A + A^\dagger) + \tfrac{1}{2}(A - A^\dagger) ;$$

here $\frac{1}{2}(A + A^\dagger)$ is self-adjoint and $\frac{1}{2}(A - A^\dagger)$ is anti-self-adjoint. We may also decompose A as follows:

$$A \equiv [\tfrac{1}{2}(A + A^\dagger)] + i[1/2\, i(A - A^\dagger)] . \tag{4.26}$$

Then both $B \equiv \frac{1}{2}(A + A^\dagger)$ and $C \equiv 1/2i(A - A^\dagger)$ are *Hermitian*. We note that B and C commute if and only if A and $A^\dagger$ commute, that is, A is normal. This result for operators is analogous to the decomposition of a complex number into the sum of a real and an imaginary part.

4.7 DIAGONALIZATION

Diagonalizing a matrix is equivalent to solving for its eigenvalues—they will simply be the diagonal elements. Thus the solution to a physical problem often involves diagonalizing a matrix. In classical mechanics, we may want to diagonalize the moment-of-inertia tensor, Eq. (1.98), and in quantum theory if we can diagonalize the matrix representation of an operator corresponding to an observable, then the diagonal elements will be the possible results of a measurement of the observable.

We have seen that the n-eigenvectors of an $n \times n$ normal matrix A with a nondegenerate spectrum (i.e., n with distinct eigenvalues) are orthogonal. Thus they are linearly independent and span the space, so by Theorem 3.16 we may construct a diagonalizing matrix P by taking as the ith column of P the ith eigenvector. Then

$$(P^{-1}AP)_{ij} = \lambda_i \delta_{ij} .$$

It is easy to see that the diagonalizing matrix P is in fact a unitary matrix. To show this, let

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & & p_{2n} \\ \vdots & & & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix} .$$

In the notation of Theorem 3.16, $X_i = (p_{1i}, \cdots p_{ni})$ is the ith eigenvector. We are free to assume it has been normalized. Since we know the eigenvectors are orthogonal, we have

$$(X_i, X_j) = \sum_{k=1}^{n} p_{ki}^* p_{kj} = \delta_{ij} ,$$

which, written in matrix notation, is just $P^\dagger P = I$, so P is unitary.

The fact that a normal matrix with distinct eigenvalues can be diagonalized by a unitary similarity transformation, is a very important result. But the restriction to nondegenerate spectra is a serious one, because most physical situations involve degenerate spectra. It would therefore be useful to remove that restriction; and there are good reasons to suspect that it can in fact be removed altogether, leaving the rest of the statement intact.

Consider the situation for Hermitian matrices. If the eigenvalues of a Hermitian matrix A are not distinct, we can perturb the elements of A by adding a Hermitian matrix δA so that the resulting Hermitian matrix $A + \delta A$ has distinct eigenvalues. This is a familiar situation in quantum theory, where the inclusion of a previously neglected perturbation, such as a magnetic field, removes the degeneracy of the spectrum. Once the degeneracy is removed, all the eigenvalues are distinct, the previous reasoning applies, and we know there exists a unitary similarity transformation that diagonalizes the matrix $A + \delta A$. As long as δA is nonzero and the eigenvalues are distinct, we know that the corresponding eigenvectors remain orthogonal, since $A + \delta A$ is *Hermitian*. (For an arbitrary matrix, this argument would break down.) Since the eigenvectors must be orthogonal for arbitrarily small but nonzero δA, it is very plausible that when δA is finally reduced to zero, they do not suddenly become linearly dependent. It is therefore reasonable to expect that as $\delta A \to 0$ we continue to have a full slate of k linearly independent eigenvectors for a k-fold root of a characteristic polynomial, and hence that there exists a unitary similarity transformation that diagonalizes A.*

We now prove that a unitary diagonalizing matrix can be constructed for any Hermitian matrix, repeated eigenvalues or not.

Theorem 4.20. Any Hermitian matrix A may be diagonalized by a unitary similarity transformation; i.e., there exists a unitary matrix U such that $U^{-1}AU$ is a diagonal matrix (the diagonal elements of which are the eigenvalues of A).

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* From a physical point of view, we know that if the perturbation δA is turned off slowly and gently, that is, *adiabatically*, the quantum numbers will not change; they are so-called *adiabatic invariants*. (In fact, it was precisely this observation that led Bohr and Sommerfeld to quantize the classical action integrals, which were known to be adiabatic invariants.) The fact that the quantum numbers do not change means that the eigenvectors remain unchanged as the perturbation diminishes adiabatically in strength. Thus there is no danger of the eigenvectors, which are orthogonal as long as $\delta A \neq 0$, becoming linearly dependent as $\delta A \to 0$.

Proof. We shall prove the theorem by induction from the 2×2 case. That is, we prove it explicitly for the 2×2 case, and then assuming it for the $(n-1) \times (n-1)$ case we show that it must hold for the $n \times n$ case. This procedure proves the theorem for any n. We could, of course, prove the theorem by induction from the 1×1 Hermitian matrix (i.e., a matrix consisting of a single real element). We use the 2×2 matrix because this is the smallest matrix for which an eigenvalue can have multiplicity greater than 1, and thus this case is more instructive.

We know the theorem is true if the eigenvalues are distinct and, in particular, it is true in this case for 2×2 matrices. In the degenerate 2×2 case, where the eigenvalues are not distinct, they must be equal. Clearly, this can be the case only if A is (already) a diagonal matrix of the form

$$A = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}.$$

(If the reader wants to verify this explicitly, he may do so by solving the characteristic polynomial of the most general 2×2 Hermitian matrix,

$$A = \begin{bmatrix} a & b \\ b^* & c \end{bmatrix},$$

when a and c are real.) In the degenerate 2×2 case, any vector is an eigenvector because

$$\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = a \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Thus we are free to choose two orthonormal eigenvectors, say

$$u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{and} \quad u_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

and therefore

$$U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

is a unitary operator which "brings" A to diagonal form by a similarity transformation, (Of course, A is in diagonal form to start with.) This exhausts all the possibilities for the 2×2 case. Now we prove the $n \times n$ case by induction.

We assume that any $(n-1) \times (n-1)$ Hermitian matrix can be diagonalized by a unitary similarity transformation. Let λ_1 be any eigenvalue of A, and denote a corresponding normalized eigenvector by

$$v_1 = (v_{11}, v_{21}, \cdots, v_{n1}).$$

We now form a unitary operator V by choosing the eigenvector v_1 as the first column of V, and putting in any other set of $n-1$ orthonormal vectors which are all orthogonal to v_1 as the remaining $n-1$ columns. Thus

$$V = (v_{i1} \mid v_{i2} \mid \cdots \mid v_{in}),$$

where the v_{ij} form the jth column vector, where $Av_1 = \lambda_1 v_1$ (that is, $\sum_l a_{kl} v_{l1} = \lambda_1 v_{k1}$), and where $(v_i, v_j) = \delta_{ij}$.

Now let us calculate the matrix $V^{-1}AV$. The general element is given by

$$(V^{-1}AV)_{ij} = (V^\dagger AV)_{ij} = \sum_k v_{ki}^* \sum_l a_{kl} v_{lj}\,, \tag{4.27}$$

and, in particular,

$$(V^{-1}AV)_{i1} = \sum_k v_{ki}^* \sum_l a_{kl} v_{l1} = \sum_k v_{ki}^* \lambda_1 v_{k1} = \lambda_1 \delta_{i1}\,.$$

Since

$$(V^{-1}AV)^\dagger = V^\dagger A^\dagger (V^{-1})^\dagger = V^{-1}AV\,,$$

$V^{-1}AV$ is Hermitian, so

$$(V^{-1}AV)_{1i} = (\lambda_1 \delta_{1i})^* = \lambda_1 \delta_{1i}\,,$$

since the eigenvalues are real. Consequently, the matrix $V^{-1}AV$ has the form

$$V^{-1}AV = \begin{bmatrix} \lambda_1 & 0 & 0 \cdots 0 \\ 0 & & \\ 0 & & \\ \vdots & & A' \\ 0 & & \end{bmatrix} \equiv B\,,$$

where A' is the $(n-1) \times (n-1)$ Hermitian matrix whose elements are given by Eq. (4.27). Now by our induction hypothesis, A' *can* be diagonalized by an $(n-1) \times (n-1)$ unitary matrix, V', that is, $(V')^{-1}A'V'$ is diagonal. Therefore, the $n \times n$ matrix $B = V^{-1}AV$ can be diagonalized.

The only remaining question is whether there is a single unitary diagonalizing matrix for A. To complete the construction explicitly, we define the unitary matrix

$$W = \begin{bmatrix} 1 & 0 \cdots 0 \\ 0 & \\ \vdots & V' \\ 0 & \end{bmatrix}.$$

Then, because $(V')^{-1}A'V'$ is diagonal (by hypothesis),

$$W^{-1}BW = W^\dagger BW$$

will be diagonal. But

$$W^{-1}BW = W^{-1}V^{-1}AVW = (VW)^{-1}A(VW) = U^{-1}AU\,,$$

where $U = VW$. The unitarity of U follows from the unitarity of V and W, as

$$U^\dagger = (VW)^\dagger = W^\dagger V^\dagger = W^{-1}V^{-1} = (VW)^{-1} = U^{-1}\,.$$

Thus there exists a unitary matrix U, which we have constructed explicitly, which diagonalizes any Hermitian matrix by a similarity transformation. The proof is complete.

This means that one can always find a complete set of orthonormal basis vectors with respect to which the matrix of a Hermitian linear transformation A is diagonal. This proof also shows that for every degenerate eigenvalue λ_0, of algebraic multiplicity $k(k \leq n)$, it is always possible to find k linearly independent eigenvectors belonging to λ_0. This follows either by directly investigating the process of construction used in the theorem, or by simply noting that the determinant of a unitary matrix always has absolute value one and so is never zero. Therefore its column vectors, which are the eigenvectors of A, must be linearly independent by property (8) of determinants (Section 3.8). Thus the algebraic and geometric multiplicities are equal for Hermitian operators; the eigenvectors of a Hermitian operator form a complete set.

This theorem has been stated for Hermitian matrices and *unitary* similarity transformations. However, as inspection will verify, the proof may be modified (by merely omitting the complex conjugates and replacing daggers by tildes) to give the following theorem.

Theorem 4.21. Any (real) symmetric matrix may be diagonalized by a real orthogonal similarity transformation.

It has been pointed out that any matrix A can be written as the sum of two Hermitian matrices: $A = B + iC$ (Eq. 4.26). We have just proved that any Hermitian matrix can be diagonalized by a unitary similarity transformation. Applying this result to B and C, we may be inclined to ask: Why can't we prove that any matrix (A), Hermitian or not, can be so diagonalized ? The answer is that diagonalizing B involves a specific set of basis vectors—eigenvectors of B—and there is no reason to believe that that the basis in which B assumes diagonal form will simultaneously cast C into diagonal form, or, in other words, that the eigenvectors of B will also be eigenvectors of C. However, if B and C do possess a common set of eigenvectors, then it will be possible to diagonalize B and C simultaneously by a single similarity transformation, and therefore A will be diagonalizable. This prompts us to look for a condition on two Hermitian matrices which ensures that they can be simultaneously diagonalized. The following theorem, which is also of great importance in quantum theory, provides such a condition.

Theorem 4.22. Two Hermitian matrices, B and C, commute if and only if there exists a complete orthonormal set of common eigenvectors.

Proof.

1. Assume that there exists a common complete orthonormal set of eigenvectors, $\{x_i\}$. Then $Bx_i = b_i x_i$ and $Cx_i = c_i x_i$. Therefore, $BCx_i = B(c_i x_i) = c_i Bx_i = c_i b_i x_i$, and similarly, $CBx_i = c_i b_i x_i$. Therefore $(BC - CB)x_i = 0$, for any x_i. But the x_i span the vector space, so $(BC - CB)x = 0$ for any x, $BC = CB$.

2. Now for the more difficult "half" of the theorem: Assume that $BC = CB$ and $Bx_i = b_i x_i$, where the $\{x_i\}$ are orthonormal. Then

$$CBx_i = b_i Cx_i \Longrightarrow B(Cx_i) = b_i(Cx_i)$$

which implies that Cx_i is an eigenvector of B corresponding to the eigenvalue b_i. There are now two possibilities:

i) The spectrum is nondegenerate, so the multiplicity of all the b_i is one. Then for all i, Cx_i must equal some constant times x_i: $Cx_i = c_i x_i$, that is, x_i is an eigenvector of C with eigenvalue c_i so the proof is complete in the nondegenerate case.

ii) The spectrum is degenerate, i.e., there exist eigenvalues b_i of multiplicity $m > 1$. Then there must exist m eigenvectors, $(Cx_i^{(1)}, Cx_i^{(2)}, \cdots, Cx_i^{(k)}, \cdots, Cx_i^{(m)})$, belonging to the eigenvalue b_i, each of which must be a linear combination of the m linearly independent eigenvectors $(x_i^{(1)}, \cdots, x_i^{(k)}, \cdots, x_i^{(m)})$ which also belong to the eigenvalue b_i. We shall choose the $x_i^{(k)}$ to be an orthonormal set, and write these linear combinations as

$$\begin{aligned} Cx_i^{(1)} &= \sum_{j=1}^{m} \alpha_{j1} x_i^{(j)} , \\ &\vdots \\ Cx_i^{(k)} &= \sum_{j=1}^{m} \alpha_{jk} x_i^{(j)} , \\ &\vdots \\ Cx_i^{(m)} &= \sum_{j=1}^{m} \alpha_{jm} x_i^{(j)} . \end{aligned} \tag{4.28}$$

The coefficient matrix (α_{ij}) is Hermitian, as the reader may show.

For the nondegenerate case, $m = 1$ for all i. Thus the treatment of the degenerate case includes the nondegenerate case. Now we want to find linear combinations of the eigenvectors, $x_i^{(k)}$, of B belonging to b_i that are simultaneously eigenvectors of C. If the linear combination

$$\sum_{k=1}^{m} \gamma_k x_i^{(k)} ,$$

were to be an eigenvector of C (as it is of B), it would be necessary that the expression

$$C\left(\sum_{k=1}^{m} \gamma_k x_i^{(k)}\right) \equiv \sum_{j,k=1}^{m} \gamma_k \alpha_{jk} x_i^{(j)}$$

be of the form

$$c\left(\sum_{j=1}^{m} \gamma_j x_i^{(j)}\right) .$$

This will be possible if there exists γ_k's such that

$$\sum_{k=1}^{m} \alpha_{jk}\gamma_k = c\gamma_j \,. \tag{4.29}$$

But this is just the eigenvalue-eigenvector problem for the $m \times m$ Hermitian matrix (α_{jk}). Hence we know there exist m linearly independent solutions (the eigenvectors) which can be chosen to be orthonormal. Thus we find m eigenvalues, c_μ, some of which may be equal, and m orthonormal eigenvectors, $\gamma^{(\mu)}$, and we have the m linear combinations

$$y_i^{(\mu)} = \sum_{k=1}^{m} \gamma_k^{(\mu)} x_i^{(k)} \,,$$

which are simultaneous orthonormal eigenvectors of B and C, as the reader may verify. QED

In quantum mechanics, this theorem (really the infinite-dimensional generalization of this theorem) shows that the operators H (Hamiltonian), L^2 (total angular momentum squared) and L_z (z-component of angular momentum), which all commute, possess a mutual set of eigenvectors and therefore the corresponding observables can be measured simultaneously. It also follows that p_x and x cannot be measured simultaneously.

Although we have proved this theorem for two *Hermitian* matrices, because of their greater relevance to quantum theory, it obviously holds for (real) symmetric matrices as well. We simply omit the complex conjugates and substitute "symmetric" for "Hermitian."

We now generalize the earlier result on the diagonalizability of Hermitian matrices (Theorem 4.20), and at the same time characterize precisely the class of matrices that can be diagonalized by a unitary similarity transformation.

Theorem 4.23. A matrix A can be diagonalized by a unitary similarity transformation if and only if it is normal.

Proof. Any matrix A can be decomposed into the sum of two Hermitian matrices; according to Eq. (4.26)

$$A = B + iC \,,$$

where B and C are Hermitian. Now A can be diagonalized by a unitary similarity transformation if and only if the Hermitian matrices B and C can be simultaneously so diagonalized; and a necessary and sufficient condition for this is that B and C commute (Theorem 4.22). But, as was mentioned in connection with Eq. (4.26), B and C commute if and only if A and $A^\dagger$ commute, i.e., if and only if A is normal. Therefore a necessary and sufficient condition for a matrix to be diagonalizable by a unitary similarity transformation is that it be a normal matrix. QED

This theorem tells us that any unitary matrix can be diagonalized by a unitary similarity transformation. A real orthogonal matrix is, of course, a

unitary matrix and therefore it can be diagonalized by a unitary similarity transformation. Note, however, that the diagonalizing matrix will, in general, be a *unitary*, not a real orthogonal matrix. For example, the rotation matrix is a real orthogonal matrix, but it can only be diagonalized by a unitary matrix.

This is to be contrasted with the situation for symmetric matrices. Since a symmetric matrix is Hermitian, it can be diagonalized by a unitary matrix. But since the eigenvalues of a symmetric matrix are real, the diagonalizing unitary matrix is in fact real orthogonal. However, the eigenvalues of a real orthogonal matrix will, in general, be complex and therefore the unitary matrix that diagonalizes it cannot usually be chosen to be real orthogonal.

4.8 ON THE SOLVABILITY OF LINEAR EQUATIONS*

In the past few sections we discussed self-adjoint operators in considerable detail. In this section, we want to obtain some results which relate the properties of a linear transformation A to the properties of the adjoint operator $A^\dagger$. These results will be used in the theory of integral equations (Chapter 8). We begin by noting the simple result:

Theorem 4.24. If A is a linear operator on a finite-dimensional vector space with eigenvalues λ_i, then $A^\dagger$ has eigenvalues λ_i^*.

Proof. If λ belongs to the spectrum of A, then $\det(A - \lambda I) = 0$. From the general relation

$$\det[(A - \lambda I)^\dagger] = [\det(A - \lambda I)]^* ,$$

it follows immediately that

$$\det(A^\dagger - \lambda^* I) = 0 ,$$

which is the statement that λ^* is in the spectrum of $A^\dagger$. QED

A more interesting result concerns the orthogonality properties of the eigenvectors of A and $A^\dagger$. The following theorem gives us an important generalization of Theorem 4.18.

Theorem 4.25. If $\{\lambda_i\}$ is the spectrum of A (and therefore $\{\lambda_i^*\}$ is the spectrum of $A^\dagger$), then if ϕ_j is any eigenfunction of A belonging to λ_j and if ψ_i is any eigenfunction of $A^\dagger$ belonging to λ_i^*, we have $(\psi_i, \phi_j) = 0$ whenever $\lambda_i \neq \lambda_j$.

Proof. Consider $A\phi_j = \lambda_j\phi_j$. Taking the inner product from the left by ψ_i, we get

$$(\psi_i, A\phi_j) = \lambda_j(\psi_i, \phi_j) ,$$

and hence

$$(A^\dagger\psi_i, \phi_j) - \lambda_j(\psi_i, \phi_j) = 0 .$$

* The material in this section will not be required until we reach Chapter 8.

But $A^\dagger\psi_i = \lambda_i^*\psi_i$, so

$$(\lambda_i - \lambda_j)(\psi_i, \phi_j) = 0 .$$

If $\lambda_i \neq \lambda_j$, we conclude that $(\psi_i, \phi_j) = 0$, as stated in the theorem. QED

Now, for a self-adjoint matrix A, we were able to show that every eigenvalue of the spectrum of A which had algebraic multiplicity $m \geq 1$ had precisely m associated linearly independent eigenvectors. For a general matrix, this result is not true, but there is a related theorem.

Theorem 4.26. Let A be a linear operator on a finite-dimensional vector space,and let λ be an eigenvalue of A with λ^* the corresponding eigenvalue of $A^\dagger$. Then if there are μ linearly independent eigenvectors, ϕ_i belonging to λ, there are also μ linearly independent eigenvectors ψ_i belonging to λ^*. Note that μ may be less than m, the algebraic multiplicity of λ (and of λ^*).

Proof. First of all, the set $\{\phi_i\}$ may be chosen to be orthonormal; if it had not been so, we could have applied the Gram-Schmidt process to it, using the fact that if ϕ_1 and ϕ_2 are both eigenvectors belonging to λ, then $\alpha\phi_1 + \beta\phi_2$ also belongs to λ. The same comment applies to the set $\{\psi_i\}$. We therefore assume that we are dealing with *orthonormal* sets. Call μ the number of elements of $\{\phi_i\}$ and ν the number of elements of $\{\psi_i\}$. We want to show that $\mu = \nu$.

We assume first that $\mu < \nu$, and define an operator B by

$$Bx \equiv Ax + \sum_{j=1}^{\mu} \psi_j(\phi_j, x) .$$

B is clearly a linear operator. We now show that λ is not an eigenvalue of B. Consider

$$Bx = \lambda x ,$$

that is,

$$Ax + \sum_{j=1}^{\mu} \psi_j(\phi_j, x) = \lambda x . \tag{4.30}$$

Taking the inner product from the left by ψ_k, we get

$$(\psi_k, Ax) + \sum_{j=1}^{\mu} (\psi_k, \psi_j)(\phi_j, x) = \lambda(\psi_k, x) .$$

But $(\psi_k, \psi_j) = \delta_{kj}$, so

$$(\psi_k, Ax) + (\phi_k, x) = \lambda(\psi_k, x) , \qquad k = 1, 2, \cdots, \mu$$

or

$$(\phi_k, x) = \lambda(\psi_k, x) - (A^\dagger\psi_k, x) .$$

However, $A^\dagger\phi_k = \lambda^*\phi_k$, so

$$(\phi_k, x) = \lambda(\phi_k, x) - (\lambda^*\phi_k, x) = 0\,, \qquad k = 1, 2, \cdots, \mu\,. \tag{4.31}$$

Returning to Eq. (4.30), we see that

$$Ax = \lambda x\,.$$

But since x is an eigenvector of A with eigenvalue λ, it can be written as

$$x = \sum_{l=1}^{\mu} c_l\phi_l\,.$$

Using Eq. (4.31) together with the orthonormality of ϕ_l, we get

$$c_k = (\phi_k, x) = 0\,, \qquad k = 1, 2, \cdots, \mu\,.$$

Thus $x = 0$. Since we have shown that $(B - \lambda I)x = 0$ implies that $x = 0$, then by Theorem 3.7 the inverse of $(B - \lambda I)$ exists and, in particular, we can solve the inhomogeneous equation

$$(B - \lambda I)x = \psi_\nu\,,$$

which can be rewritten as

$$Ax + \sum_{j=1}^{\mu} \psi_j(\phi_j, x) - \lambda x = \psi_\nu\,.$$

We now take the inner product from the left by ψ_ν and find

$$(\psi_\nu, Ax) - \lambda(\psi_\nu, x) = 1\,,$$

where the term involving the summation vanishes because of assumption $\nu > \mu$. Since $A^\dagger\psi_\nu = \lambda^*\psi_\nu$, we conclude that

$$(\lambda^*\psi_\nu, x) - \lambda(\psi_\nu, x) = 1$$

or $0 = 1$, a contradiction. This impossibility leads us to the conclusion that our original assumption $(\nu > \mu)$ is inconsistent. Hence $\mu \geq \nu$. The same argument, using the operator C defined by

$$Cx \equiv A^\dagger x + \sum_{j=1}^{\nu} \phi_j(\psi_j, x)\,,$$

shows that $\mu > \nu$ is also impossible, so we are left with just $\mu = \nu$. QED

We emphasize again that even though $\mu = \nu$, μ may nevertheless be smaller (but not larger) than m, the algebraic multiplicity of λ and λ^*. With this result, we can obtain an important generalization of Theorem 3.17.

Theorem 4.27. In a finite-dimensional vector space, the equation

$$(A - \lambda I)x = y\,,$$

has a solution if and only if $(\psi_i, y) = 0$ for all ψ_i satisfying

$$(A^\dagger - \lambda^* I)\psi_i = 0.$$

Proof. The desired result follows readily from the previous theorem. There we established that if

$$A\phi_i = \lambda\phi_i \qquad (i = 1, 2, \cdots, \mu)$$

and

$$A^\dagger\psi_i = \lambda^*\psi_i \qquad (i = 1, 2, \cdots, \mu) ,$$

then λ is not an eigenvalue of B, where B is defined by

$$B \equiv Ax + \sum_{j=1}^{\mu} \psi_j(\phi_j, x) .$$

Hence the equation

$$(B - \lambda I)x = \left[Ax + \sum_{j=1}^{\mu} \psi_j(\phi_j, x) - \lambda x\right] = y \tag{4.32}$$

has a solution because $\det(B - \lambda I) \neq 0$ and $(B - \lambda I)^{-1}$ therefore exists. Let us first assume that $(\psi_i, y) = 0$ for all i. Then taking the inner product from the left with ψ_i in Eq. (4.32), we find that

$$(\psi_i, Ax) + (\phi_i, x) - \lambda(\psi_i, x) = (\psi_i, y) , \qquad i = 1, 2, \cdots, \mu .$$

Since $(\psi_i, Ax) = (A^\dagger\psi_i, x) = (\lambda^*\psi_i, x) = \lambda(\psi_i, x)$, it follows that

$$(\phi_i, x) = (\psi_i, y) = 0 \qquad \text{for all } i.$$

Hence Eq. (4.32) reduces to

$$(A - \lambda I)x = y . \tag{4.33}$$

Since Eq. (4.32) has a solution, so does Eq. (4.33), which is what was to be shown.

The converse is trivial to prove. If we assume that $(A - \lambda I)x = y$ has a solution x, then taking the inner product on the left by ψ_i, we get

$$(\psi_i, Ax) - \lambda(\psi_i, x) = (\psi_i, y) , \qquad i = 1, 2, \cdots, \mu .$$

Since we just saw that $(\psi_i, Ax) = \lambda(\psi_i, x)$, we have immediately $(\psi_i, y) = 0$, for all i. QED

Note that if we assume that A has a complete set of eigenvectors, then the above theorem reduces to the results of Theorem 3.17. For in this case, calling $\{x_i\}$ the eigenvectors of A, we write

$$x = \sum_{i=1}^{n} \xi_i x_i , \qquad y = \sum_{i=1}^{n} \eta_i x_i ,$$

so

$$(A - \lambda I)x = \sum_{i=1}^{n} (\lambda_i - \lambda)\xi_i x_i = y = \sum_{i=1}^{n} \eta_i x_i ,$$

where λ_i is the eigenvalue corresponding to x_i. Thus

$$\sum_{i=1}^{n} [(\lambda_i - \lambda)\xi_i - \eta_i]x_i = 0,$$

and since the x_i are linearly independent,

$$(\lambda_i - \lambda)\xi_i = \eta_i,$$

for all i. If $\lambda_\nu = \lambda$, then a solution can exist only if $\eta_\nu = 0$. Similarly, if a solution does exist when $\lambda_\nu = \lambda$, then $\eta_\nu = 0$. Clearly, if $\lambda_\nu = \lambda$ and $\eta_\nu = 0$, then coefficient ξ_ν is completely arbitrary. Assuming for notational convenience that only one λ_ν gives trouble, the solution for x is

$$x = \sum_{\substack{i=1 \\ i \neq \nu}}^{n} \frac{\eta_i}{\lambda_i - \lambda} x_i + a x_\nu,$$

where a is *any* constant. Thus, under the assumption of a complete set of eigenvectors belonging to A, we can obtain an explicit solution to the equations, as we did in Theorem 3.17, where we took $\lambda = 0$ for convenience. Note that the relation $\eta_\nu = 0$ means that y has the form

$$y = \sum_{\substack{i=1 \\ i \neq \nu}}^{n} \eta_i x_i.$$

But, denoting by y_ν the eigenvector of $A^\dagger$ belonging to λ_ν^*, we have by Theorem 4.25,

$$(y_\nu, y) = 0,$$

so that the statement $\eta_\nu = 0$ is equivalent to $(y_\nu, y) = 0$. Again we are assuming a nondegenerate spectrum. The wording can easily be changed slightly to cover the case of a multiple eigenvalue.

Thus with virtually no assumptions about the operator A, we have been able to obtain very general conditions for the solvability of a system of linear equations. These results will be of great importance when we discuss integral equations in Chapter 8.

4.9 MINIMUM PRINCIPLES

In this section we want to show how the concepts of the calculus of variations can be used to obtain some striking results when applied to inner products in a vector space.

In Chapter 2, we considered the general problem of extremizing the value of an integral with some fairly smooth function in the integrand. The Euler-Lagrange equations that we found there are second-order differential equations which are, in general, nonlinear and therefore very complicated. Consider, for

instance, a basic problem of classical physics—Newton's second law for an arbitrary potential:

$$m\ddot{\mathbf{r}} = -\nabla V(\mathbf{r}) .$$

As we saw in Section 2.5, this equation can be derived readily from the Euler-Lagrange equations for the system. But it is, in general, a system of nonlinear equations which is extremely difficult to solve. This difficulty is often concealed by the fact that special forms for $V(\mathbf{r})$ lead to the existence of conserved quantities. In such cases, the solution of this system of second-order differential equations is reduced to doing the first integrals.

Now there is a special, but nevertheless important, class of problems for which the variational technique is useful and can also be put in a much more general mathematical setting than we have heretofore considered. Let us suppose that we have a finite- or infinite-dimensional inner product space V, and let A be some *linear* operator mapping V into itself. Now consider the quantity

$$I = (x, Ax) .$$

The vector x depends continuously on some parameter ϵ in such a way that the inner product is a function of ϵ. Then we write

$$I(\epsilon) = (x(\epsilon), Ax(\epsilon)) .$$

We are interested in extremizing $I(\epsilon)$ under certain conditions. Therefore we shall want to take derivatives of $I(\epsilon)$. If we assume that A is a linear operator and that $x(\epsilon)$ is a differentiable function of ϵ, then it is easy to establish the "chain rule":

$$\frac{dI}{d\epsilon} = \left(\frac{dx}{d\epsilon}, Ax\right) + \left(x, A\frac{dx}{d\epsilon}\right) . \tag{4.34}$$

We obtain this simple result because A is a *linear* operator. Note that x is a very general object. It could be an n-tuple of numbers, or it could be a function of several variables (besides ϵ) in a space of functions with an inner product given, for example, by an integral over the space.

Equation (4.34) suggests that the techniques of Chapter 2 might be applicable to the problem of extremizing inner products on a vector space. This is indeed true if A is *Hermitian*. Let us now consider this case in detail. We want to extremize the expression

$$I = (x, Ax) ,$$

where A is self-adjoint and $x \in V$, an inner-product space. Now the reader will see immediately that we have not posed the problem in quite the right way. For, given any x which leads to, say, a negative value of I, we can produce an arbitrarily large negative value of I just by increasing the norm of x. Similarly, if we have an x which gives some positive value for I, then we can produce an arbitrarily large positive value of I by increasing the norm of x. Thus $I = 0$ is the only possible stationary value for the problem as we have stated it. To prevent this artificial increase in magnitude of I, let us *constrain* the vectors x

to have a fixed norm. For convenience, we shall take $||x|| = 1$. Then rephrased, the problem is to extremize I with respect to the class of vectors x satisfying the constraint equation

$$J = (x, x) = 1 .$$

To handle this problem, we introduce a two-parameter family of vectors,* $\bar{x}(\epsilon_1, \epsilon_2)$, satisfying

a) $\bar{x}(\epsilon_1, \epsilon_2) \in V$ for all ϵ_1, ϵ_2;

b) $\bar{x}(0, 0) = x$, the extremizing vector;

c) $\partial\bar{x}/\partial\epsilon_1$ and $\partial\bar{x}/\partial\epsilon_2$ exist and are in V.

Then we form

$$I(\epsilon_1, \epsilon_2) = (\bar{x}(\epsilon_1, \epsilon_2), A\bar{x}(\epsilon_1, \epsilon_2)) , \tag{4.35a}$$

$$J(\epsilon_1, \epsilon_2) = (\bar{x}(\epsilon_1, \epsilon_2), \bar{x}(\epsilon_1, \epsilon_2)) = 1 , \tag{4.35b}$$

and consider

$$K(\epsilon_1, \epsilon_2) = I(\epsilon_1, \epsilon_2) - \lambda J(\epsilon_1, \epsilon_2) , \tag{4.35c}$$

where λ is a Lagrange multiplier. According to condition (b) above,

$$\left.\frac{\partial K}{\partial \epsilon_1}\right]_{\epsilon_1=\epsilon_2=0} = 0 = \left.\frac{\partial K}{\partial \epsilon_2}\right]_{\epsilon_1=\epsilon_2=0} \tag{4.36}$$

is the condition for an extremum. Note that Eqs. (4.35 a, b, c) tell us that

$$K(\epsilon_1, \epsilon_2) = (\bar{x}(\epsilon_1, \epsilon_2), [A - \lambda]\bar{x}(\epsilon_1, \epsilon_2)) .$$

We write $(A - \lambda I)$ as simply $(A - \lambda)$ to economize. Also, since A is Hermitian, $I(\epsilon_1, \epsilon_2)$ is real, so λ is real. We now use Eq. (4.34) to calculate the two first partial derivatives of K.

$$\frac{\partial K}{\partial \epsilon_1} = \left(\frac{\partial \bar{x}}{\partial \epsilon_1}, [A - \lambda]\bar{x}\right) + \left(\bar{x}, [A - \lambda]\frac{\partial \bar{x}}{\partial \epsilon_1}\right) ,$$

$$\frac{\partial K}{\partial \epsilon_2} = \left(\frac{\partial \bar{x}}{\partial \epsilon_2}, [A - \lambda]\bar{x}\right) + \left(\bar{x}, [A - \lambda]\frac{\partial \bar{x}}{\partial \epsilon_2}\right) .$$

Setting $\epsilon_1 = 0 = \epsilon_2$ and using Eq. (4.34), we find that

$$(\eta_1, [A - \lambda]x) + (x, [A - \lambda]\eta_1) = 0 , \tag{4.37a}$$

$$(\eta_2, [A - \lambda]x) + (x, [A - \lambda]\eta_2) = 0 , \tag{4.37b}$$

where

$$\eta_1 = \left.\frac{\partial \bar{x}}{\partial \epsilon_1}\right]_{\epsilon_1=\epsilon_2=0} , \qquad \eta_2 = \left.\frac{\partial \bar{x}}{\partial \epsilon_2}\right]_{\epsilon_1=\epsilon_2=0} ;$$

* Note that the argument does not depend on the number of parameters (except that *at least* two are needed). In practice one can use more than two parameters if necessary.

$(A - \lambda)$ is self-adjoint, because λ is real. Hence, from Eqs. (4.37a, b),

$$([A - \lambda]x, \eta_i) + (\eta_i, [A - \lambda]x) = 0 , \qquad i = 1, 2.$$

Now η_i is arbitrary, so we may set $\eta_i = (A - \lambda)x$. We conclude that

$$2([A - \lambda]x, [A - \lambda]x) = 0 , \qquad \text{so} \qquad (A - \lambda)x = 0 .$$

We have, therefore, a remarkable link between the eigenvalue problem for self-adjoint operators and the calculus of variations, which can be stated as follows:

Theorem 4.28. If A is a linear self-adjoint operator, $Ax = \lambda x$ if and only if $I = (x, Ax)$ is extremized with respect to the constraint $(x, x) = 1$. The extremized value of I is just λ, the Lagrange multiplier.

It is interesting to see under what circumstances I actually takes on a minimum value. First, it is clear that if λ is any eigenvalue other than the lowest one, the extremum of I will not be a minimum. Therefore only the lowest eigenvalue and its corresponding eigenvector can yield a minimum for I.

In a finite-dimensional vector space, there is always a lowest eigenvalue. In this case, we have the theorems which establish completeness for self-adjoint operators (Theorems 4.20 and 4.21) at our disposal. Thus, any $x \in V$ can be written as

$$x = \sum_{i=1}^{n} \alpha_i x_i ,$$

where the x_i are the eigenfunctions of A and where we take the α_i to be real for convenience. (The reader can easily go through the discussion with α_i complex, handling the real and imaginary parts as independent variables). Thus calling λ_i the eigenvalues of A and ordering them according to $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$, we find that

$$(x, Ax) = \sum_{i=1}^{n} \alpha_i^2 \lambda_i , \qquad (x, x) = \sum_{i=1}^{n} \alpha_i^2 = 1 .$$

We have already proved that in any inner-product space, the eigenvalues provide us with an extremum of (x, Ax). Therefore, we need not show this again in the case of a finite-dimensional space. However, in order to illustrate the ideas involved, we derive this result once more for this simple case. In trying to extremize (x, Ax) with respect to α_i, holding $(x, x) = 1$, we set the first derivatives of

$$K(\alpha_i) = \sum_{i=1}^{n} \alpha_i^2 (\lambda_i - \lambda)$$

equal to zero, where λ is a Lagrange multiplier. We have

$$\frac{dK}{d\alpha_i} = 2\alpha_i(\lambda_i - \lambda) , \qquad i = 1, \cdots, n .$$

Thus the extrema are determined by

$$\alpha_i(\lambda_i - \lambda) = 0\,, \qquad i = 1, \cdots, n\,.$$

If λ is not equal to any of the λ_i, then the only possible solution is $\alpha_i = 0$ for all i. This is not admissible, since then

$$\sum_{i=1}^{n} \alpha_i^2 \neq 1\,.$$

Therefore, for some j, with $1 \leq j \leq n$, $\lambda = \lambda_j$; also, $\alpha_j = 1$ and $\alpha_i = 0$ for $i \neq j$.

Now if $\lambda = \lambda_1$, $\alpha_1 = 1$, and $\alpha_i = 0$ for $i \geq 2$. Then all the second derivatives of K with respect to the α_i, for $i \geq 2$, are positive and I takes on a minimum in this case. The value of I at this minimum is precisely λ_1. For an extremum at $\lambda = \lambda_k$, $1 < k < n$, $\alpha_k = 1$, and $\alpha_i = 0$ for $i \neq k$. Now some of the second partial derivatives will be positive and some negative, so we have a saddle point of some type. For the case $\lambda = \lambda_n$, all the second derivatives are negative and we have a maximum; this maximum value of I is just $I = \lambda_n$. Hence we obtain

Theorem 4.29. Let A be a self-adjoint operator in an n-dimensional inner-product space. If λ_1 is the least eigenvalue of A and λ_n the largest eigenvalue then the minimum and maximum values of (x, Ax) are λ_1 and λ_n, respectively, for all vectors $x \in V$ such that $(x, x) = 1$.

We shall return to the infinite-dimensional extension of this theorem in later chapters. The reader may have noticed that we also assumed tacitly above that the spectrum of A was nondegenerate. A little work will show that this assumption is unnecessary.

Now consider λ_2, the second smallest eigenvalue of A. If we take the class of all x such that $(x, x) = 1$, then λ_2 is a saddle point of (x, Ax), not a true minimum. However, if we restrict ourselves to the smaller class of vectors satisfying both $(x, x) = 1$ and $(x, x_1) = 0$ where x_1 is the eigenvector belonging to λ_1, then (x, Ax) takes on its smallest value when $x = x_2$, and this smallest value is λ_2. If we restrict the normalized vector x to lie entirely within the subspace of V spanned by $\{x_2, x_3, \cdots, x_n\}$, or better, entirely outside the subspace of V spanned by x_1, then (x, Ax) takes on its minimum value for $x = x_2$ and this minimum value is just λ_2.

A definition will be useful in formulating the final generalization of these considerations.

Definition 4.9. If V is a vector space and U is a subspace of V, the set of all vectors *not* lying in U is called the *complement of U in V*; we denote this set by U_c.

Then we have

Theorem 4.30. Let A be a self-adjoint operator in an n-dimensional inner-product space V, whose eigenvalues are ordered by $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$.

Further, let V_i be the subspace of V spanned by the eigenvectors of A belonging to the first i eigenvalues. Then for the class of vectors $\{x\}$ which are normalized to unity and lie in the complement of V_i, the minimum value of (x, Ax) with respect to members of this class is assumed when $x = x_{i+1}$, and the minimum value is λ_{i+1}.

Proof. The result follows immediately if we note that all x in the class can be written in the form

$$x = \sum_{j=i+1}^{n} \alpha_j x_j$$

so that

$$(x, Ax) = \sum_{j=i+1}^{n} \alpha_j^2 \lambda_j \qquad \text{and} \qquad (x, x) = \sum_{j=i+1}^{n} \alpha_j^2 = 1 .$$

From this point on, the proof proceeds in the same fashion as the proof of the previous theorem. QED

As we have derived them, Theorems 4.29 and 4.30 are valid only for finite-dimensional spaces. However, Theorem 4.28 is valid for a space of any dimension, and this suggests that, since the eigenfunctions of A produce extrema of (x, Ax), it may be possible to find minimum theorems similar to 4.29 and 4.30 for infinite-dimensional spaces. We shall find in Chapter 9 that this is indeed the case.

Now the difficulty with which we must come to grips in the infinite-dimensional case is twofold:

1. there may be an infinite number of eigenvectors;
2. there may be a continuous spectrum.

A glance at the proofs of Theorems 4.29 and 4.30 suggests that difficulty (1) is not serious. However, difficulty (2) is very serious indeed. In fact, it will force us to revise our concept of completeness, upon which the proofs of Theorems 4.29 and 4.30 are based.

It turns out that for most Hermitian operators of interest in physics, (x, Ax) takes on a minimum value equal to the lowest eigenvalue of A when x is equal to the corresponding eigenvector. In most physical problems, it is intuitively clear that a lowest eigenvalue exists. For example, in the nonrelativistic Schrodinger equation for a potential of given strength, it is obvious that the binding energy due to that potential cannot be an arbitrarily large negative number. Thus we suspect that there will be a lower bound for the energy eigenvalues. However, all this discussion of minimum principles will be irrelevant if there is no such lowest eigenvalue.

On the other hand, a particle not in a bound state can have an arbitrarily large positive energy, so we might guess that the spectrum resulting from the Schrodinger equation in this case will not be bounded above. Moreover, we know on physical grounds that these so-called *scattering states* can have any

energy whatever, as opposed to bound states, which are restricted to certain discrete energy levels, as in the case of the hydrogen atom. Therefore, this part of the spectrum corresponds to states which can lie arbitrarily close together. This is roughly what we mean by the term *continuous spectrum*.

A slightly more mathematical way of looking at the continuous spectrum is to say that it corresponds to functions which "solve" a given operator equation but do not lie in the vector space. We shall go into such matters in more detail in later chapters, but we can suggest some of the difficulties of dealing with infinite-dimensional spaces by considering the quantum-mechanical kinetic-energy operator

$$\frac{p^2}{2m} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} \tag{4.38}$$

acting on the vector space of functions f which satisfy the condition

$$(f, f) = \int_{-\infty}^{\infty} |f|^2\, dx < \infty\,.$$

Such functions are said to be *square-integrable*; we shall show later that they do indeed form a vector space.

Now consider the eigenvalue equation

$$\frac{p^2}{2m}\phi = E\phi\,. \tag{4.39}$$

Clearly, $\phi = e^{ikx}$, $E = \hbar^2k^2/2m$ are solutions to the differential equation implied by Eqs. (4.38) and (4.39), for *all* k, and indeed they are the only solutions if E is positive. However,

$$\int_{-\infty}^{\infty} |\phi|^2\, dx = \infty$$

for every value of k, so the ϕ's do not lie in the vector space in question. Nevertheless, it turns out that every function in the space can be written as a linear combination of the ϕ's in the generalized sense that, if $f(x)$ is in the space of square-integrable functions, there exists a function $\tilde{f}(k)$ such that

$$f(x) = \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx}\, dk.$$

This is the famous Fourier integral theorem. It assures us that every vector in the space of square-integrable functions can be written as a *continuous* linear combination of the "eigenvector" of Eq. (4.39).

Thus the eigenvectors for a continuous eigenvalue spectrum look as though they can be handled very much like vectors in finite-dimensional vector spaces —provided that the notion of completeness can be generalized in a suitable way. The study of such problems is central in the theory of infinite-dimensional vector spaces.

Some of the most interesting and relevant applications of the theorems of this section are in the solution of eigenvalue problems for infinite-dimensional

vector spaces. Ignoring for now the fact that we have not rigorously established these results for such spaces, let us consider some examples from quantum mechanics.

Let A be a given linear operator. One possible technique for finding the lowest eigenvalue is to take a trial vector x with a number of variable parameters, calculate (x, Ax), and then minimize this inner product with respect to the variable parameters in x. By adding more and more parameters, we can make the value of (x, Ax) smaller and smaller; however, it can never drop below λ_1, the lowest eigenvalue of A. Clearly, we must be guided by common sense in this process; for example, if we were inept enough to pick a trial vector in such a way that it was always orthogonal to the true eigenvector belonging to the lowest eigenvalue, then by Theorem 4.30, we would actually be closing in on the second eigenvalue, λ_2, as we increased the number of parameters. A consideration of the physics of the problem will usually enable us to avoid such pitfalls.

Examples. The Hamiltonian of the quantum harmonic oscillator is

$$H = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 x^2 .$$

If we choose p and x as in Eq. (3.39), this becomes

$$H = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2} m\omega^2 x^2 .$$

Let us call our trial vector ϕ and consider

$$I = (\phi, H\phi) = \int_{-\infty}^{\infty} \phi^*(x) H\phi(x)\, dx .$$

Now, on physical grounds alone we would expect the true eigenfunction to fall off rather rapidly with distance because of the strong harmonic attraction. Hence, a reasonable guess for ϕ might be

$$\phi = Ce^{-\alpha|x|/2}, \qquad \alpha > 0 .$$

The constraint $(\phi, \phi) = 1 = C^2 \int_{-\infty}^{\infty} e^{-\alpha|x|}\, dx$ can be met by choosing $C = (\alpha/2)^{1/2}$. Thus

$$\phi(x) = (\sqrt{\alpha/2})e^{-\alpha|x|/2}, \qquad \alpha > 0 ,$$

and putting this value of ϕ into our expression for I, we find that

$$I = (\hbar^2\alpha^2/8m) + (m\omega^2/\alpha^2) .$$

It is evident that the extremum for I is a minimum; also, a Lagrange multiplier is not necessary since we have normalized ϕ to 1. Thus, setting $dI/d\alpha = 0$, we find that the minimum occurs at

$$\alpha = \alpha_0 = \left(\frac{8m^2\omega^2}{\hbar^2}\right)^{1/4} .$$

The value which I takes on at this minimum is $I(\alpha_0) = (1/\sqrt{2})\hbar\omega$, which is greater than the eigenvalue, $\lambda_1 = \frac{1}{2}\hbar\omega$, by about 40 percent. We have seen in Section 3.10 that the correct eigenvector is

$$u_0 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-(m\omega/2\hbar)x^2},$$

which does not resemble our result,

$$\phi = \left(\frac{m^2\omega^2}{2\hbar^2}\right)^{1/8} \exp\left[-\left(\frac{m^2\omega^2}{2\hbar^2}\right)^{1/4} |x|\right],$$

too closely. Nevertheless, the value which we find for energy is only 40 percent larger than the true lowest eigenvalue. An interesting problem is to guess a trial function of the form

$$\phi' = Ce^{-\alpha x^2/2}$$

and to show that upon minimizing $(\phi', H\phi')$ one obtains the true eigenvalue and eigenvector.

As another example, consider the calculation of the ground-state energy of helium. The Hamiltonian is

$$H = -\frac{\hbar^2}{2m}(\nabla_1^2 + \nabla_2^2) - 2e^2\left(\frac{1}{r_1} + \frac{1}{r_2}\right) + \frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|},$$

where 1 and 2 index the two electrons moving about the nucleus of charge $Z = +2e$. The final term in this Hamiltonian representing the Coulomb repulsion between the electrons makes the problem impossible to solve exactly. If we neglect it the problem can be solved, giving for the ground-state eigenfunction the product of two ground-state hydrogen atom eigenfunctions:

$$\phi_0 = \left(\frac{Z^3}{\pi a^3}\right) \exp\left(-\frac{Z(r_1 + r_2)}{a}\right),$$

where $a = \hbar^2/me^2 =$ the Bohr radius. The corresponding energy is -54.4 eV, which is not in particularly good agreement with the experimental value of -78.60 eV. However, if we use a trial function of this *form* and replace Z by $Z_{\text{eff}} = Z - \sigma$, we can do much better. This is physically reasonable because each electron is partially screened from the full charge of the nucleus by the presence of the other electron. We then want to minimize

$$I = (\phi_0, H\phi_0),$$

where H is the complete Hamiltonian for the helium atom, including the interaction between electrons, and

$$\phi_0 = \left[\frac{(Z - \sigma)^3}{\pi a^3}\right] \exp\left[-\frac{(Z - \sigma)(r_1 + r_2)}{a}\right].$$

We know that $I(\sigma) \geq E_0$, the ground state energy of the helium atom. By direct integration, we find that

$$I(\sigma) = -(e^2/a)(\tfrac{11}{4} + \tfrac{5}{8}\sigma - \sigma^2),$$

$I(\sigma)$ has a minimum for $\sigma = \frac{5}{16}$. Therefore the best estimate of E_0 with a trial function of this form is

$$E_0 = I(\tfrac{5}{16}) = -\tfrac{729}{256}(e^2/a) = -77.46\,\text{eV}\,,$$

since $e^2/a = 27.20\,\text{eV}$. This is quite close to (and greater than) the measured value of $-78.60\,\text{eV}$.

Trial functions with many adjustable parameters can be used. They lead to phenomenally close agreement between theory and experiment.*

4.10 NORMAL MODES

Now that we know how to bring a linear operator to diagonal form, it is a simple matter to define powers of a fairly large class of linear operators. Let A be a linear operator on an n-dimensional vector space, and suppose it can be diagonalized by a similarity transformation

$$S^{-1}AS = D\,,$$

where D is defined by

$$D_{ij} = \lambda_i \delta_{ij}\,,$$

and $\lambda_1, \lambda_2, \cdots, \lambda_n$ are the eigenvalues of A. Then A^α, where α is any real number, may be defined by

$$A^\alpha = SD^\alpha S^{-1}\,, \qquad \text{where} \qquad [D^\alpha]_{ij} = \lambda_i^\alpha \delta_{ij}\,.$$

Note that if α is negative and A has a zero eigenvalue, then A^α is not well defined. Since $SS^{-1} = I$, it is easily seen that $A^\alpha A^\beta = A^{(\alpha+\beta)}$.

Self-adjoint operators can always be diagonalized, so we focus our attention on them. A particular class of self-adjoint operators for which negative powers always exist is the positive-definite operator.

Definition 4.10. An operator H on a vector space V is said to be *positive-definite* if and only if $(x, Hx) > 0$ for all nonzero x in V. If we have just $(x, Hx) \geq 0$, H is called a *positive* operator.

Clearly, H is self-adjoint since (x, Hx) is real for all x. Also, a positive-definite operator must have strictly positive eigenvalues since

$$Hx = \lambda x$$

implies that

$$\lambda = \frac{(x, Hx)}{(x, x)}\,,$$

so if $(x, Hx) > 0$ for all nonzero x, then $\lambda > 0$, since $x = 0$ can never be an eigenvector.

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* See C. L. Pekeris, *Phys. Rev.* **115**, 1216 (1959).

For positive-definite H, $H^{1/2}$ is Hermitian, which may be seen as follows:

$$H^{1/2} = UD^{1/2}U^{-1} = UD^{1/2}U^{\dagger},$$

where U is the *unitary* operator which, by Theorem 4.20, diagonalizes H. Since the eigenvalues of H are positive, $D^{1/2}$ is a real diagonal matrix, and therefore $D^{1/2}$ is Hermitian. Thus

$$(H^{1/2})^{\dagger} = (UD^{1/2}U^{\dagger})^{\dagger} = (U^{\dagger})^{\dagger}(D^{1/2})^{\dagger}U^{\dagger} = UD^{1/2}U^{\dagger} = H^{1/2},$$

so $H^{1/2}$ is self-adjoint, as is also $H^{-1/2}$. In a similar way, it follows that if H is (real) symmetric, then so are $H^{1/2}$ and $H^{-1/2}$.

A particularly interesting example of a positive-definite operator occurs in classical mechanics. Suppose that $(x_1, \cdots, x_n)$ is a system of generalized coordinates—not necessarily orthogonal—which may be considered as components of a vector x.* Under many circumstances, the kinetic energy of the system will take the form

$$K = \tfrac{1}{2} \sum_{i,j} T_{ij}\dot{x}_i\dot{x}_j,$$

where (T_{ij}) is an array of real constants, e.g., masses, which are independent of the coordinates and of the time. They may be chosen so that $T_{ij} = T_{ji}$. The kinetic energy K can be written very compactly in our notation as

$$K = \tfrac{1}{2}(\dot{x}, T\dot{x}),$$

where the components of the vector $\dot{x}$ are $\dot{x}_1, \dot{x}_2, \cdots, \dot{x}_n$. Since K is the kinetic energy, it must be strictly positive for any nonzero vector x; thus T is positive-definite as well as (real) symmetric.

The potential energy will, in general, be a complicated function of the coordinates. But if we confine our attention to small bounded motions about the equilibrium point, which we can take without loss of generality to be located at $x = 0$, then we may write

$$V(x) = [V]_{x=0} + \sum_i \left[\frac{\partial V}{\partial x_i}\right]_{x=0} x_i + \tfrac{1}{2} \sum_{i,j} \left[\frac{\partial^2 V}{\partial x_i \partial x_j}\right]_{x=0} x_i x_j + \cdots$$

The subscript on each bracket indicates that the quantity inside the bracket is evaluated at $x = 0$. Equilibrium corresponds to an extremum of the potential energy, so

$$\left[\frac{\partial V}{\partial x_i}\right]_{x=0} = 0, \qquad \text{for all } i.$$

Since the constant term can only shift the zero of energy we drop it; also, we neglect terms in x_i^3 and higher, since we are interested in small amplitude

* For reasons of convenience, we are going to suspend the notational conventions for vectors and their components throughout this section.

motion. Thus

$$V(x) = \tfrac{1}{2} \sum_{i,j} U_{ij} x_i x_j \,, \qquad \text{where} \qquad U_{ij} = \left[\frac{\partial^2 V}{\partial x_i \partial x_j}\right]_{x=0} = U_{ji} \,,$$

so U is (real) symmetric. In our more compact notation,

$$V = \tfrac{1}{2} (x, Ux) \,.$$

The matrix U is an array of *constants*, as was the matrix T in the expression for the kinetic energy. For bounded motion, we want a local minimum of the potential energy, so that if we move away from the equilibrium point in any direction the potential energy will increase. This may be described mathematically by saying

$$\sum_{i,j} \left[\frac{\partial^2 V}{\partial x_i \partial x_j}\right]_{x=0} \epsilon_i \epsilon_j > 0 \,,$$

for all possible choices of the vector ϵ. For many physical purposes, it is convenient to weaken this condition slightly and require only that

$$\sum_{i,j} \left[\frac{\partial^2 V}{\partial x_i \partial x_j}\right]_{x=0} \epsilon_i \epsilon_j \geq 0 \,,$$

for all possible ϵ. Thus U is a *positive* operator, so that

$$(x, Ux) \geq 0 \,.$$

We shall now study the general dynamical solutions for the motion of such a system, making use of the powerful machinery developed in this chapter. This will give us the main results of the theory of coupled systems without recourse to the messy algebra often encountered in traditional discussions.

For such a system, the Lagrangian may be written as

$$L = K - V = \tfrac{1}{2} (\dot{x}, T\dot{x}) - \tfrac{1}{2} (x, Ux) \,.$$

From Lagrange's equations,

$$\frac{\partial L}{\partial x_i} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}_i}\right) = 0 \qquad \text{for all } i,$$

we get immediately the set of equations

$$\sum_j T_{ij} \ddot{x}_j + \sum_j U_{ij} x_j = 0 \qquad \text{for all } i.$$

This can be written simply as

$$T\ddot{x} + Ux = 0 \,. \tag{4.40}$$

By analogy with the simple harmonic-oscillator problem, let us look for a solution of the form

$$x = \xi e^{\pm i\omega t} \,,$$

where the scalar ω and the vector ξ are to be determined. Substituting this into Eq. (4.40), we get

$$U\xi - \omega^2 T\xi = 0 , \tag{4.41}$$

which is an eigenvalue problem, but of a type we have not encountered before. Our notation suggests that ω^2 should not be negative, and this is indeed the case since

$$\omega^2 = \frac{(\xi, U\xi)}{(\xi, T\xi)} ,$$

and $(\xi, U\xi) \geq 0$, $(\xi, T\xi) > 0$.

The properties of the solutions to Eq. (4.41) are easily found by using the existence of $T^{1/2}$ and $T^{-1/2}$, although we shall see that in solving an actual problem it will not be necessary to compute $T^{1/2}$ or $T^{-1/2}$. Equation (4.41) implies that

$$T^{-1/2}U\xi = \omega^2 T^{1/2}\xi ,$$

or since $T^{-1/2}T^{1/2} = I$,

$$T^{-1/2}UT^{-1/2}T^{1/2}\xi = \omega^2 T^{1/2}\xi .$$

Thus if we call

$$\eta = T^{1/2}\xi \tag{4.42}$$

and

$$W = T^{-1/2}UT^{-1/2} ,$$

we have

$$W\eta = \omega^2\eta , \tag{4.43}$$

which is a simple eigenvalue problem of the usual form. It is clear that $T^{-1/2}$ will be (real) symmetric if T is real and positive-definite, as well as symmetric. Therefore since U is (real) symmetric,

$$\tilde{W} = \widetilde{T^{-1/2}UT^{-1/2}} = (\widetilde{T^{-1/2}})\tilde{U}(\widetilde{T^{-1/2}}) = T^{-1/2}UT^{-1/2} = W ,$$

and thus W is (real) symmetric.

Thus, by Theorem 4.21, W has n eigenvalues ω_i^2, some of which may have multiplicity greater than one, and n real orthonormal eigenvectors $\eta^{(i)}$, where

$$(\eta^{(i)}, \eta^{(j)}) = \delta_{ij} .$$

With the $\eta^{(i)}$ determined, we find the corresponding $\xi^{(i)}$ which are solutions to Eq. (4.41). Equation (4.42) yields immediately

$$\xi^{(i)} = T^{-1/2}\eta^{(i)} .$$

For the $\xi^{(i)}$, Eq. (4.42) yields

$$(\eta^{(i)}, \eta^{(j)}) = (T^{1/2}\xi^{(i)}, T^{1/2}\xi^{(j)}) = (\xi^{(i)}, T\xi^{(j)}) ,$$

so that

$$(\xi^{(i)}, T\xi^{(j)}) = \delta_{ij} . \tag{4.44}$$

Since the $\eta^{(i)}$ are real, so are the $\xi^{(i)}$. From the $\xi^{(i)}$, we get as solutions to Eq. (4.40) the vectors $x_{\pm}^{(i)}(t)$:

$$x_{\pm}^{(i)}(t) = \xi^{(i)} e^{\pm i\omega_i t} .$$

The magnitude of these vectors is as yet undetermined. In a physical problem, we would have a set of boundary conditions to determine these magnitudes, for example, $x_i(0) = \alpha_i$ and $\dot{x}_i(0) = \beta_i$ for all i, which in our notation looks like $x(0) = \alpha$, $\dot{x}(0) = \beta$. Our general solution can be written as a linear combination of the $x_{\pm}^{(i)}(t)$. Thus

$$\begin{aligned} x(t) &= \sum_k [C_k x_{+}^{(k)}(t) + D_k x_{-}^{(k)}(t) \\ &= \sum_k \xi^{(k)}[C_k e^{i\omega_k t} + D_k e^{-i\omega_k t}] . \end{aligned}$$

In order to emphasize the fact that $x(t)$ is real, we can use a linear combination of $\sin \omega_k t$ and $\cos \omega_k t$ instead of $e^{+i\omega_k t}$ and $e^{-i\omega_k t}$. We have then

$$x(t) = \sum_k \xi^{(k)}[A_k \cos \omega_k t + B_k \sin \omega_k t] . \tag{4.45}$$

Applying the boundary conditions, we obtain

$$\alpha = \sum_k \xi^{(k)} A_k , \qquad \beta = \sum_k \omega_k \xi^{(k)} B_k .$$

Hence

$$\begin{aligned} (\xi^{(j)}, T\alpha) &= \sum_k (\xi^{(j)}, T\xi^{(k)}) A_k , \\ (\xi^{(j)}, T\beta) &= \sum_k \omega_k (\xi^{(j)}, T\xi^{(k)}) B_k , \end{aligned}$$

and using the orthogonality relation of Eq. (4.44), we get

$$A_j = (\xi^{(j)}, T\alpha) , \qquad B_j = (\xi^{(j)}, T\beta)/\omega_j . \tag{4.46}$$

If the boundary conditions are such that all the A_j and B_j vanish except for one value of j, then we have simple harmonic motion with only one frequency, ω_j, occurring. The relative amplitude with which each coordinate varies under these circumstances is given by the various components of $\xi^{(j)}$. It is called the jth *normal mode*, and ω_j is called the *normal frequency* of the jth mode. The most general motion is a linear superposition of normal modes.

The meaning of those normal modes can be made clearer by looking at the Lagrangian, L, once more:

$$L = \tfrac{1}{2}(\dot{x}, T\dot{x}) - \tfrac{1}{2}(x, Ux) .$$

Writing, in analogy with Eq. (4.42),

$$y = T^{1/2}x , \qquad x = T^{-1/2}y ,$$

we get

$$L = \tfrac{1}{2}(\dot{y}, \dot{y}) - \tfrac{1}{2}(y, T^{-1/2}UT^{-1/2}y) ,$$

where we have made use of the fact that $T^{-1/2}$ is self-adjoint. Using our definition W in Eq. (4.42), we get

$$L = \tfrac{1}{2}(\dot{y}, \dot{y}) - \tfrac{1}{2}(y, Wy) .$$

However, our previous work tells us that W is real symmetric and therefore can be diagonalized by an orthogonal transformation O. The diagonal matrix obtained by the similarity transformation of W with O is a matrix Ω with the square of the normal frequencies down the diagonal; O is just an orthogonal matrix whose columns are the $\eta^{(j)}$. Since $\tilde{O}O = I$, we get

$$\begin{aligned} L &= \tfrac{1}{2}(\dot{y}, O\tilde{O}\dot{y}) - \tfrac{1}{2}(y, O\tilde{O}WO\tilde{O}y) \\ &= \tfrac{1}{2}(\tilde{O}\dot{y}, \tilde{O}\dot{y}) - \tfrac{1}{2}(\tilde{O}y, \Omega\tilde{O}y) . \end{aligned}$$

If we call $z = \tilde{O}y$, then

$$L = \tfrac{1}{2}(\dot{z}, \dot{z}) - \tfrac{1}{2}(z, \Omega z) . \tag{4.47}$$

Thus, in terms of this specially constructed vector z, the Lagrangian assumes a simple form which is most obvious if L is written in terms of components:

$$L = \tfrac{1}{2}\sum_{i=1}^{n} (\dot{z}_i^2 - \omega_i^2 z_i^2) ,$$

whose equations of motion are just

$$\ddot{z}_i + \omega_i^2 z_i = 0 \qquad \text{for all } i.$$

Therefore, by going to this coordinate system, we find that the coupled oscillatory motion decouples into a set of simple harmonic motions whose solutions are

$$z_i = A_i \cos \omega_i t + B_i \sin \omega_i t \qquad \text{for all } i.$$

These various z_i correspond to the normal modes discussed above. This may be seen as follows. With z determined, we can invert our previous steps to get $x = T^{-1/2}Oz$. We leave it to the reader to show that the matrix O' defined by $O' = T^{-1/2}O$ is just the matrix whose columns are composed of the vectors $\xi^{(j)}$:

$$O'_{ik} = \xi_i^{(k)} . \tag{4.48}$$

O' is not an orthogonal matrix, although it is easy to show that $\tilde{O}'TO' = I$. Thus we have $x = O'z$, and if we write this in components, we get

$$x_i = \sum_k O'_{ik} z_k \qquad \text{for all } i.$$

Using Eq. (4.48), we obtain

$$x_i = \sum_k \xi_i^{(k)} (A_k \cos \omega_k t + B_k \sin \omega_k t) ,$$

where we have made use of our general solution for z_k. In vector notation this is just

$$x = \sum_k \xi^{(k)}(A_k \cos \omega_k t + B_k \sin \omega_k t) ,$$

which is our result of Eq. (4.45). Thus we see that our definition of normal modes corresponds exactly to the solution of the transformed Lagrangian of Eq. (4.47): simple harmonic motion of one particular coordinate in a special coordinate system !

As we promised earlier, we never actually need to obtain $T^{\pm 1/2}$; we used this only to discover the general properties of the solutions. The procedure involving $T^{\pm 1/2}$ is *not* a practical algorithm. In practice, we work directly with Eq. (4.41) and solve the equation

$$\det (U - \omega^2 T) = 0$$

to determine the possible values of ω^2. Then the eigenvectors $\xi^{(i)}$ are found from Eq. (4.41). The boundary conditions are applied by using Eq. (4.46).

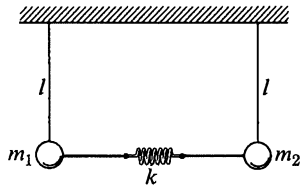

Fig. 4.1. Two pendula of length l coupled by a spring with spring constant k. m_1 and m_2 are the masses of the two bobs.

Consider, for example, the system shown in Fig. 4.1. The kinetic energy is

$$K = \tfrac{1}{2} m_1 l^2 \dot{\theta}_1^2 + \tfrac{1}{2} m_2 l^2 \dot{\theta}_2^2 ,$$

where θ_1 and θ_2 are the angular displacements of m_1 and m_2 from the vertical. Thus

$$T = \begin{bmatrix} m_1 l^2 & 0 \\ 0 & m_2 l^2 \end{bmatrix} .$$

For small oscillations,

$$V = \tfrac{1}{2} m_1 g l \theta_1^2 + \tfrac{1}{2} m_2 g l \theta_2^2 + \tfrac{1}{2} k l^2 (\theta_1 - \theta_2)^2 ,$$

so

$$U = \begin{bmatrix} m_1 g l + k l^2 & -k l^2 \\ -k l^2 & m_2 g l + k l^2 \end{bmatrix} .$$

Now

$$\det (U - \lambda T) = 0$$

implies that

$$\det\begin{bmatrix} m_1gl + kl^2 - m_1l^2\lambda & -kl^2 \\ -kl^2 & m_2gl + kl^2 - m_2l^2\lambda \end{bmatrix} = 0\,,$$

which leads to the secular equation

$$[m_1l^2(\lambda - g/l) - kl^2][m_2l^2(\lambda - g/l) - kl^2] - k^2l^4 = 0\,.$$

The solutions are

$$\lambda_1 = \frac{g}{l}\,, \qquad \lambda_2 = \frac{g}{l} + \frac{k}{\mu}\,,$$

where $\mu = m_1m_2/(m_1 + m_2)$. Thus

$$\omega_1 = \left(\frac{g}{l}\right)^{1/2}, \qquad \omega_2 = \left(\frac{g}{l} + \frac{k}{\mu}\right)^{1/2}.$$

The corresponding eigenvectors are

$$\xi^{(1)} = N\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \xi^{(2)} = M\begin{bmatrix} -m_2 \\ m_1 \end{bmatrix},$$

where N and M must be determined so that $\xi^{(1)}$ and $\xi^{(2)}$ are properly normalized. The first mode has just the frequency which one simple pendulum would have and corresponds to the two pendula swinging in phase with each other without compressing the spring. The second mode corresponds to the pendula beating against each other, thus compressing the spring. Equation (4.44) gives the normalization as

$$\xi^{(1)} = \frac{1}{l(m_1 + m_2)^{1/2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

$$\xi^{(2)} = \frac{1}{l(m_1 + m_2)^{1/2}}\begin{bmatrix} -(m_2/m_1)^{1/2} \\ +(m_1/m_2)^{1/2} \end{bmatrix}.$$

One can easily check that $(\xi^{(1)}, T\xi^{(2)}) = 0$ as required.

To find the complete solution, we need to know the initial conditions. Suppose that for the time $t = 0$, $\theta_1 = \alpha_1$, and $\theta_2 = \alpha_2$ give the positions of m_1 and m_2. Then in the notation of Eq. (4.46),

$$\theta(0) = \begin{bmatrix} \theta_1(0) \\ \theta_2(0) \end{bmatrix} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \alpha \qquad \dot\theta(0) = \begin{bmatrix} \dot\theta_1(0) \\ \dot\theta_2(0) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \beta\,.$$

Thus Eq. (4.45) tells us that $B_1 = B_2 = 0$ and

$$A_1 = (\xi^{(1)}, T\alpha) = \frac{1}{l(m_1 + m_2)^{1/2}}(1, 1)\begin{bmatrix} m_1l^2 & 0 \\ 0 & m_2l^2 \end{bmatrix}\begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \frac{\alpha_1m_1 + \alpha_2m_2}{(m_1 + m_2)^{1/2}}\,l;$$

$$A_2 = (\xi^{(2)}, T\alpha)$$

$$= \frac{1}{l(m_1 + m_2)^{1/2}}\left[-\left(\frac{m_2}{m_1}\right)^{1/2}, \left(\frac{m_1}{m_2}\right)^{1/2}\right]\begin{bmatrix} m_1l^2 & 0 \\ 0 & m_2l^2 \end{bmatrix}\begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \mu^{1/2}(\alpha_2 - \alpha_1)l\,.$$